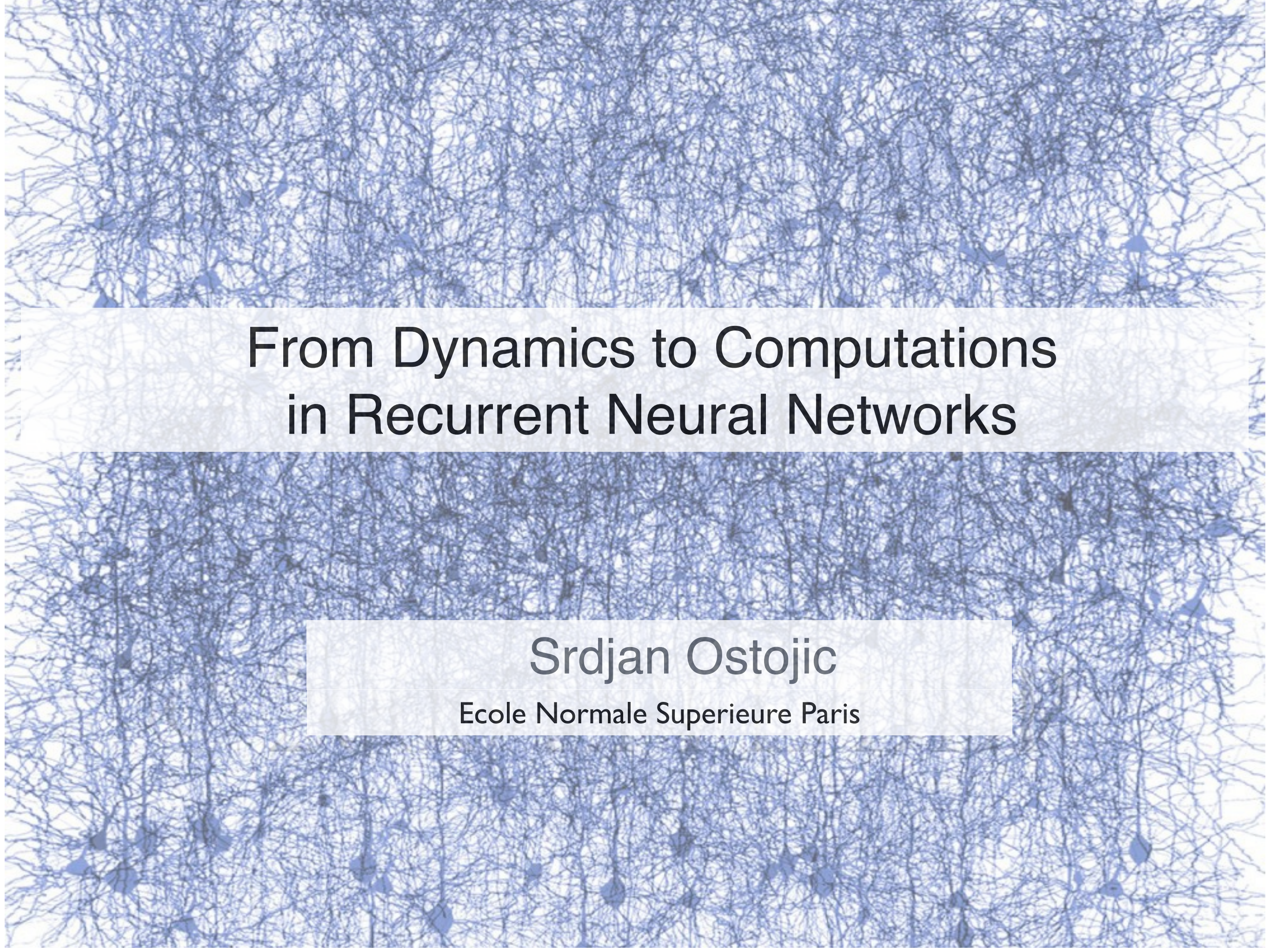


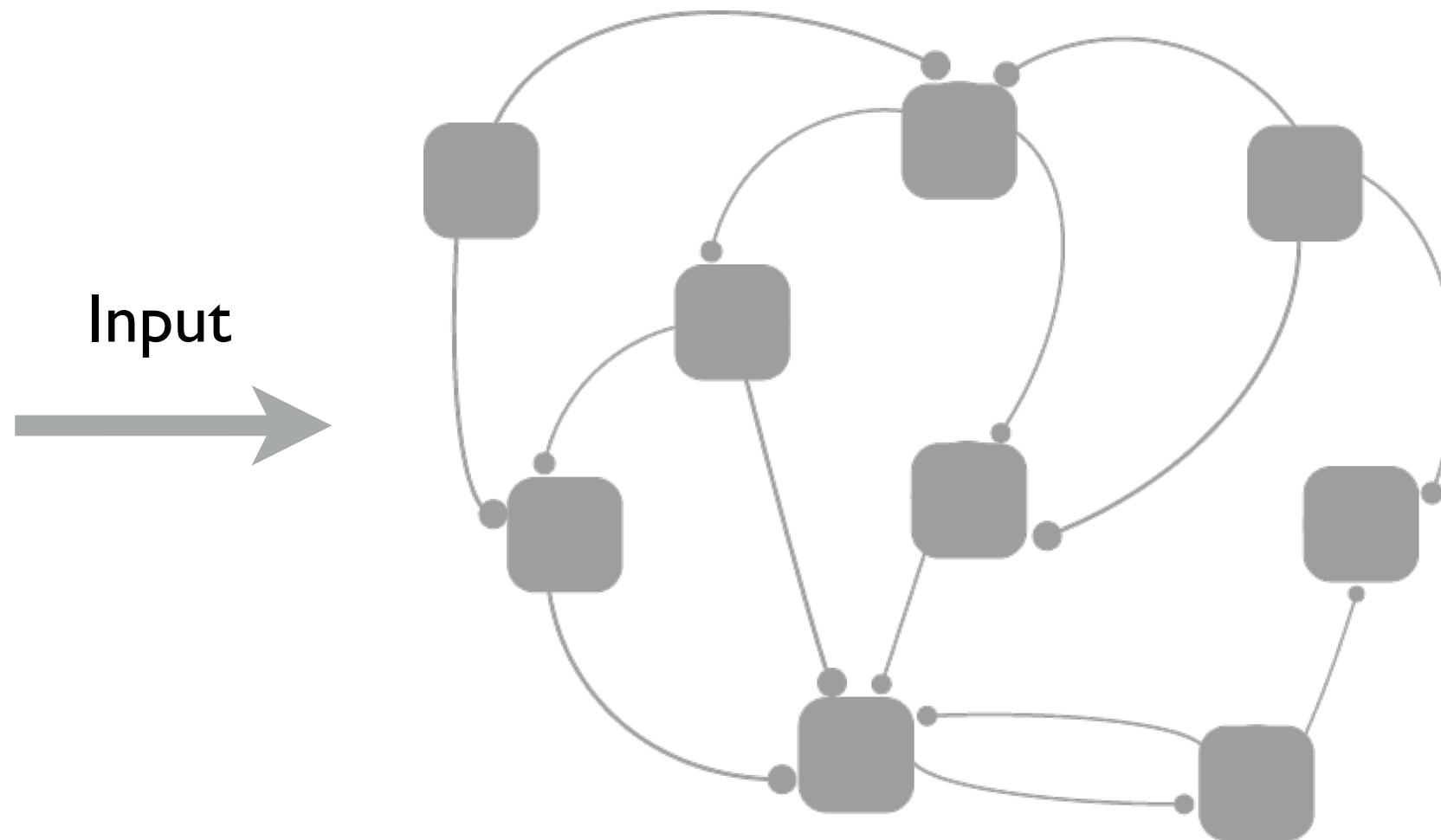


# From Dynamics to Computations in Recurrent Neural Networks

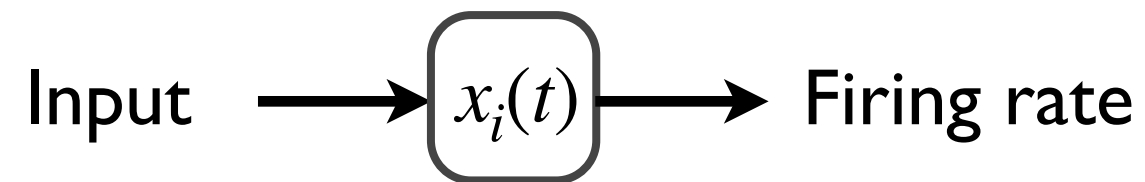
Srdjan Ostojic

Ecole Normale Supérieure Paris

# Recurrent networks



# Individual neurons: rate units



Activation of unit i:

$$\tau \frac{dx_i}{dt} = -x_i + I_i^{(tot)}(t)$$

Firing rate:

$$r_i(t) = \phi(x_i)$$

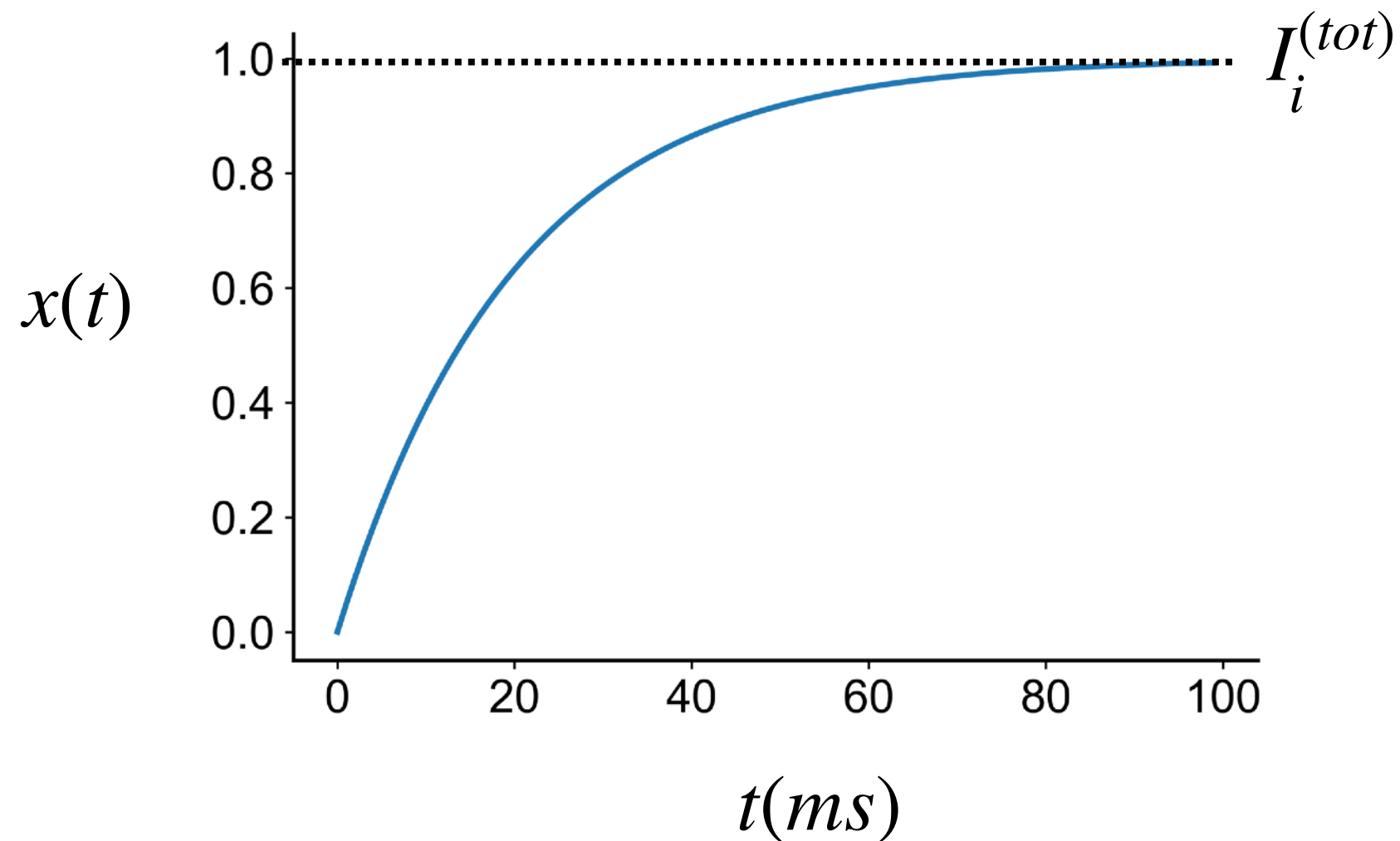
# Activation dynamics

$$\tau \frac{dx_i}{dt} = -x_i + I_i^{(tot)}$$

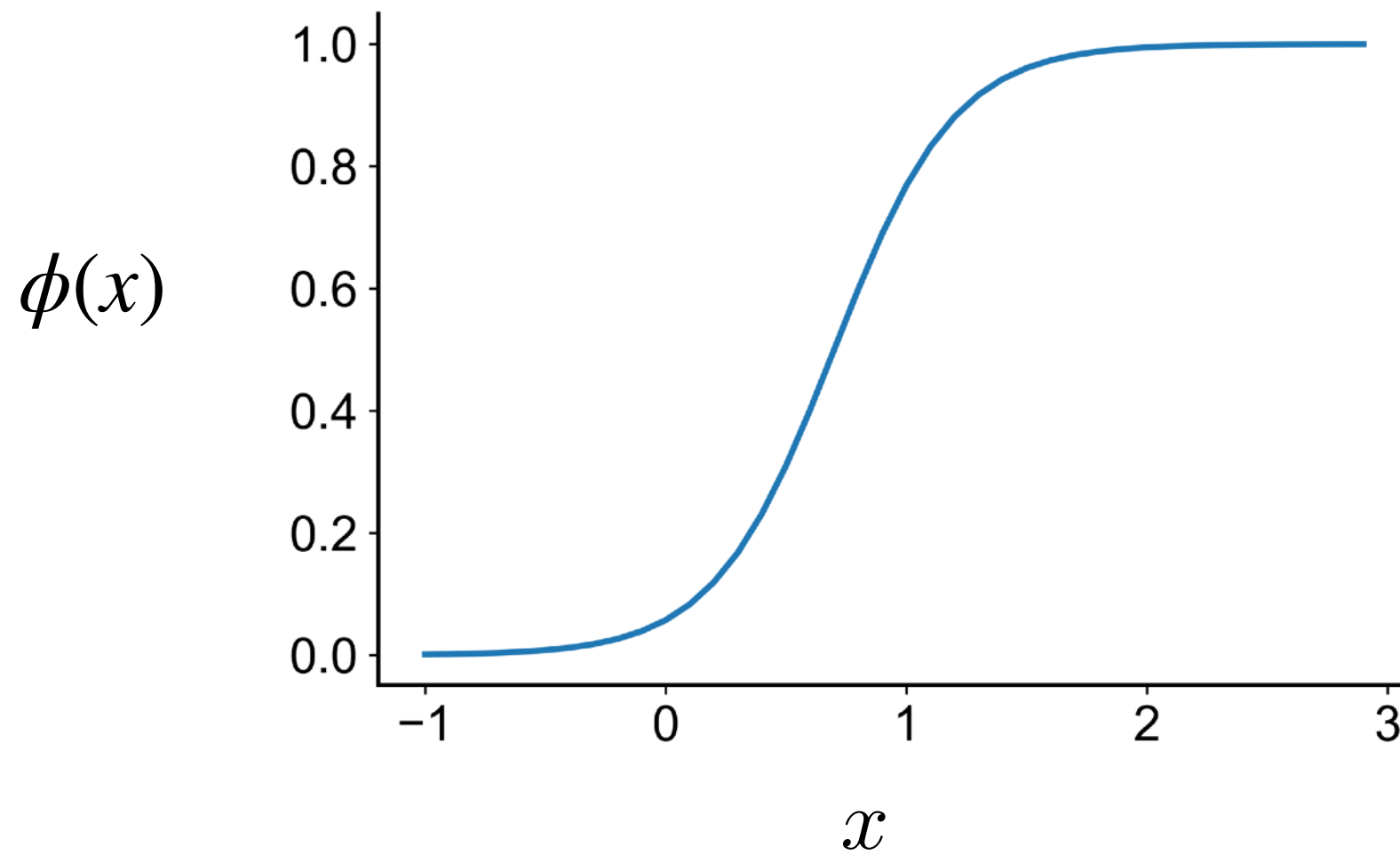
# Activation dynamics

$$\tau \frac{dx_i}{dt} = -x_i + I_i^{(tot)}$$

$$x_i(t) = I_i^{(tot)} + (x_0 - I_i^{(tot)})e^{-(t-t_0)/\tau}$$



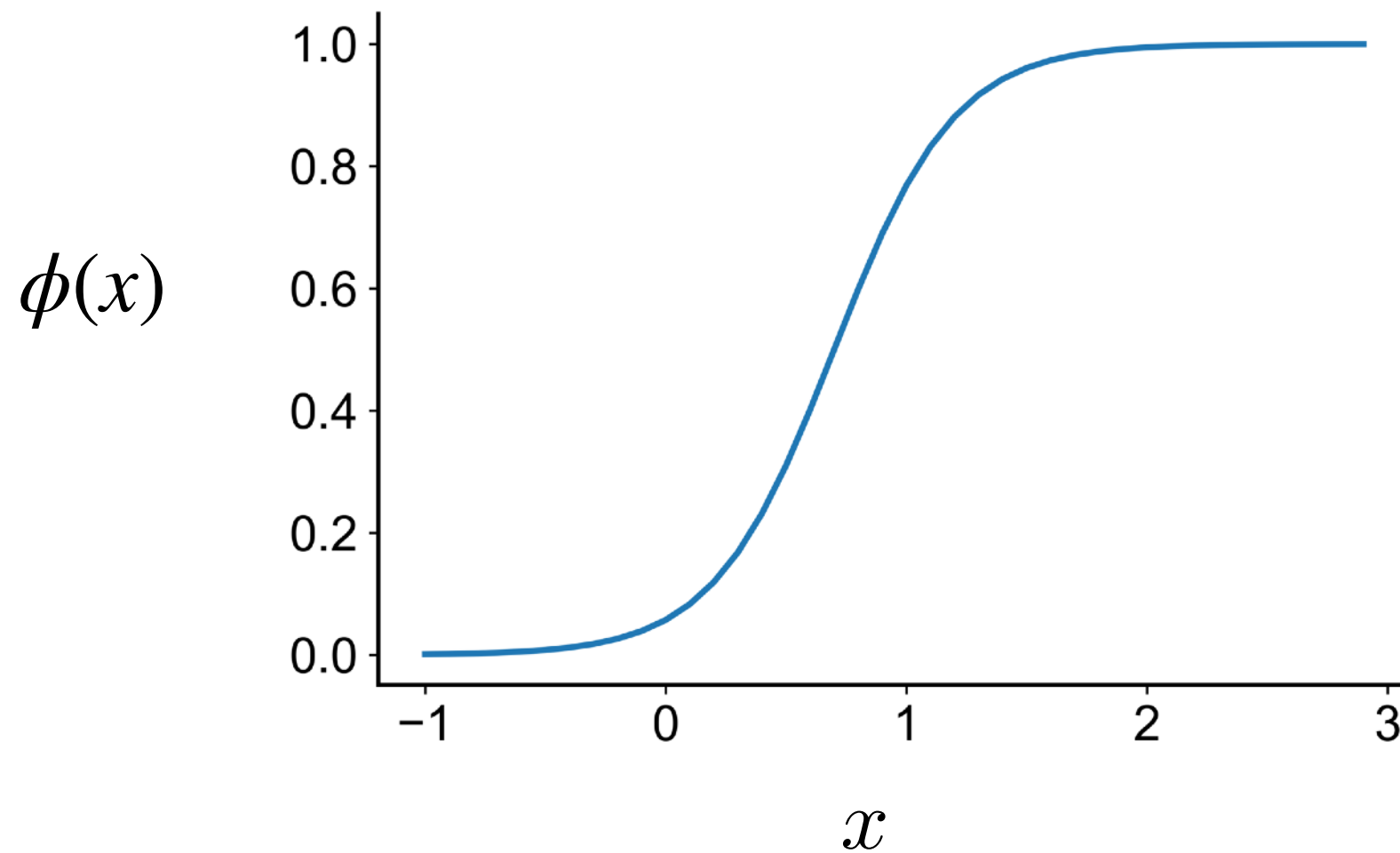
# Rate function



“sigmoid”

# Rate function

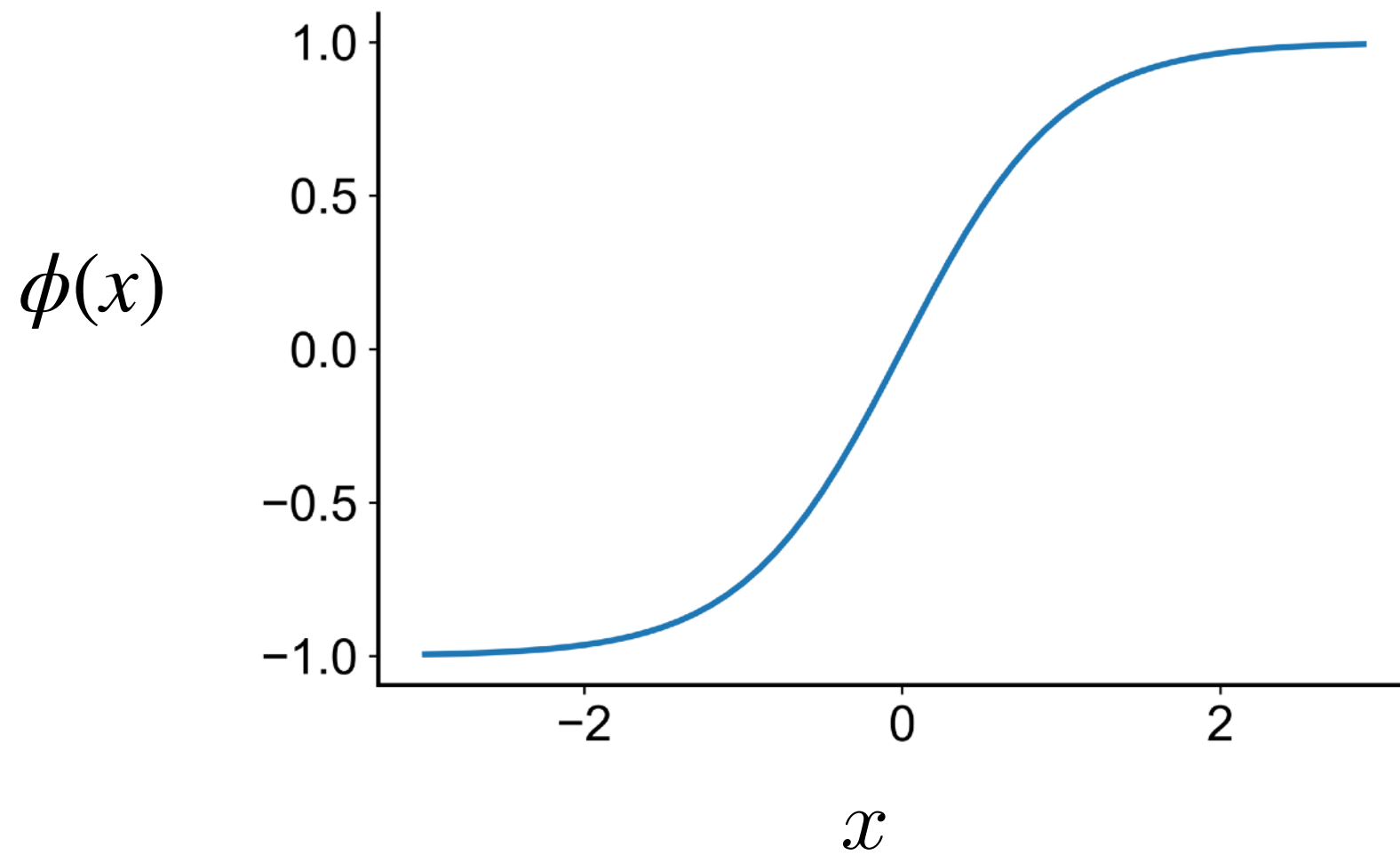
$$\phi(x) = \frac{1}{1 + e^{-\beta(x-\gamma)}}$$



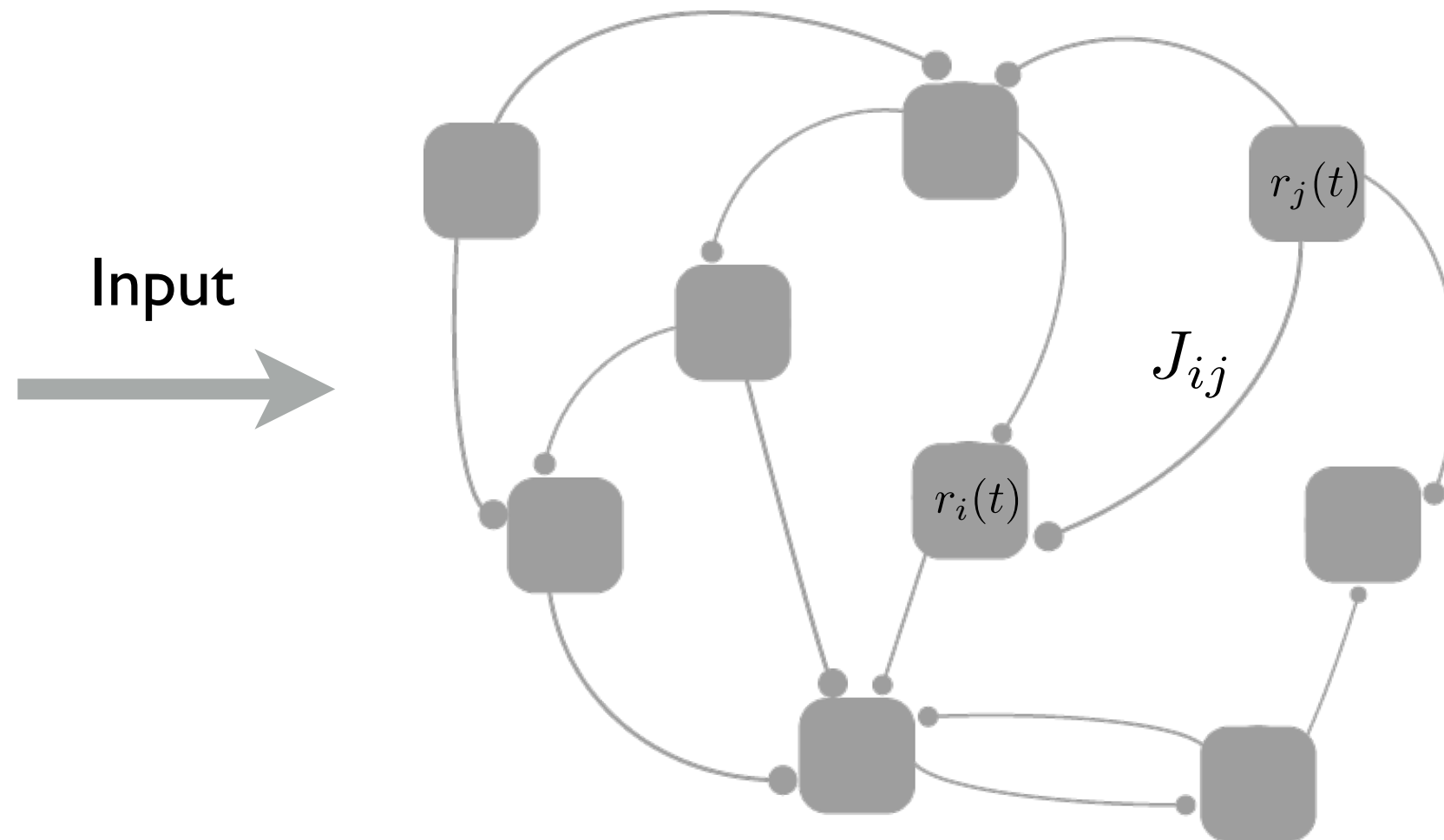
“sigmoid”

# Rate function

$$\phi(x) = \tanh(x)$$



# Recurrent networks



# Connecting different units

Input to unit  $i$  from unit  $j$ :

$$I_{j \rightarrow i} = J_{ij} r_j(t)$$

Total input to unit  $i$ :

$$I_i^{(tot)} = \sum_{j=1}^N J_{ij} r_j(t) + I_i^{(ext)}$$

# Rate networks

Activation of unit i:

$$\tau \frac{dx_i}{dt} = -x_i + \underbrace{\sum_{j=1}^N J_{ij} \phi(x_j)}_{\text{recurrent input}} + \underbrace{I_i^{(ext)}(t)}_{\text{external input}}$$

coupling strength from unit j to i

rate of unit j

→ Connectivity matrix

$r_j(t) = \phi(x_j)$

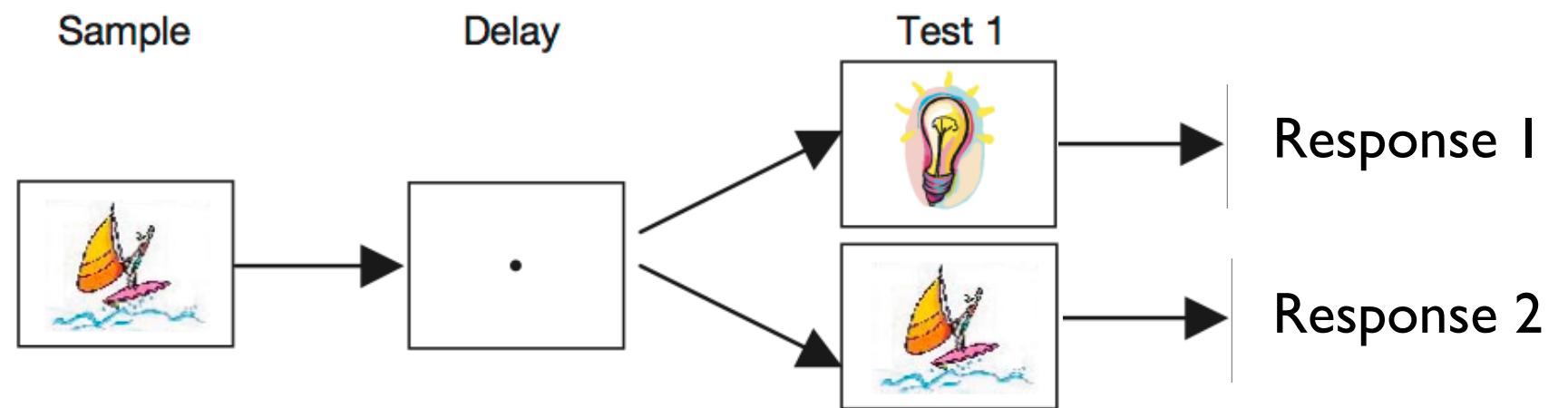
# Questions

- How does network structure determine dynamics?
- How do dynamics of neural networks lead to computations?

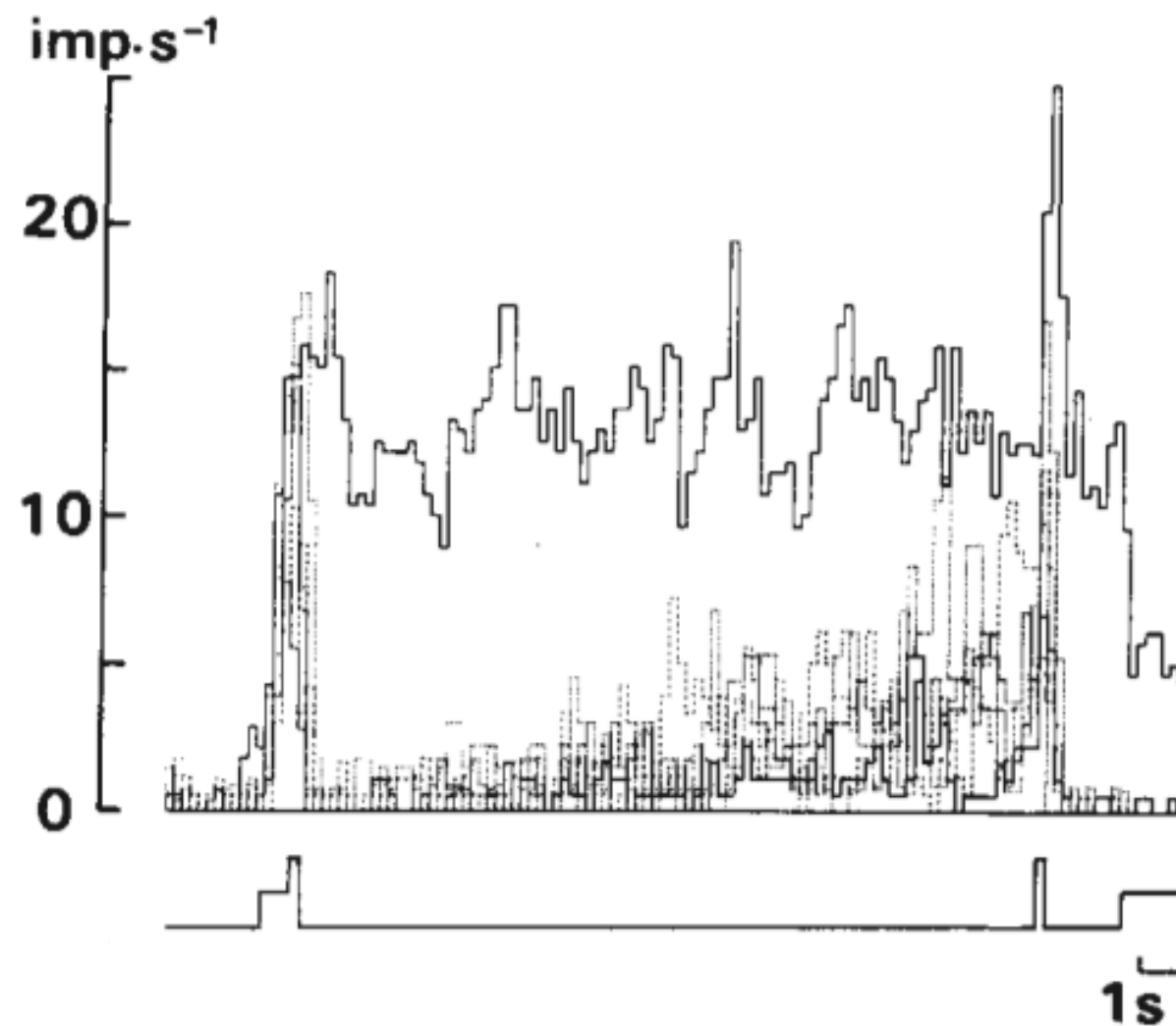
Connectivity → Dynamics → Computations

# Computations: example I

## Delayed match-to sample task



# Persistent mnemonic activity

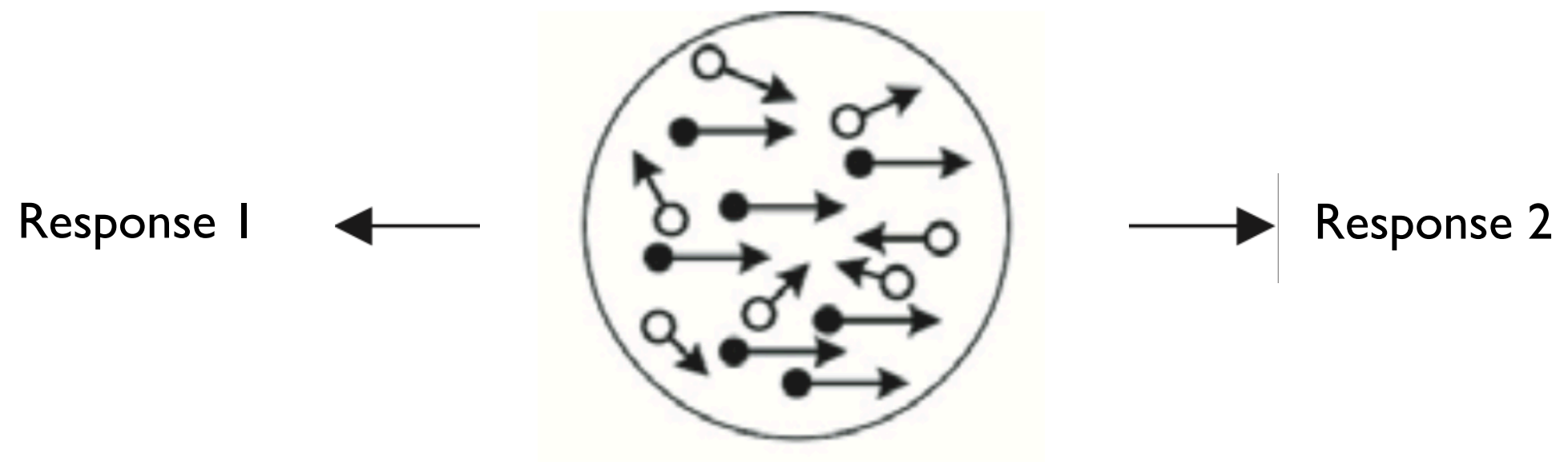


➔ Working memory encoded in persistent activity

[Miyashita and Chang 1988]

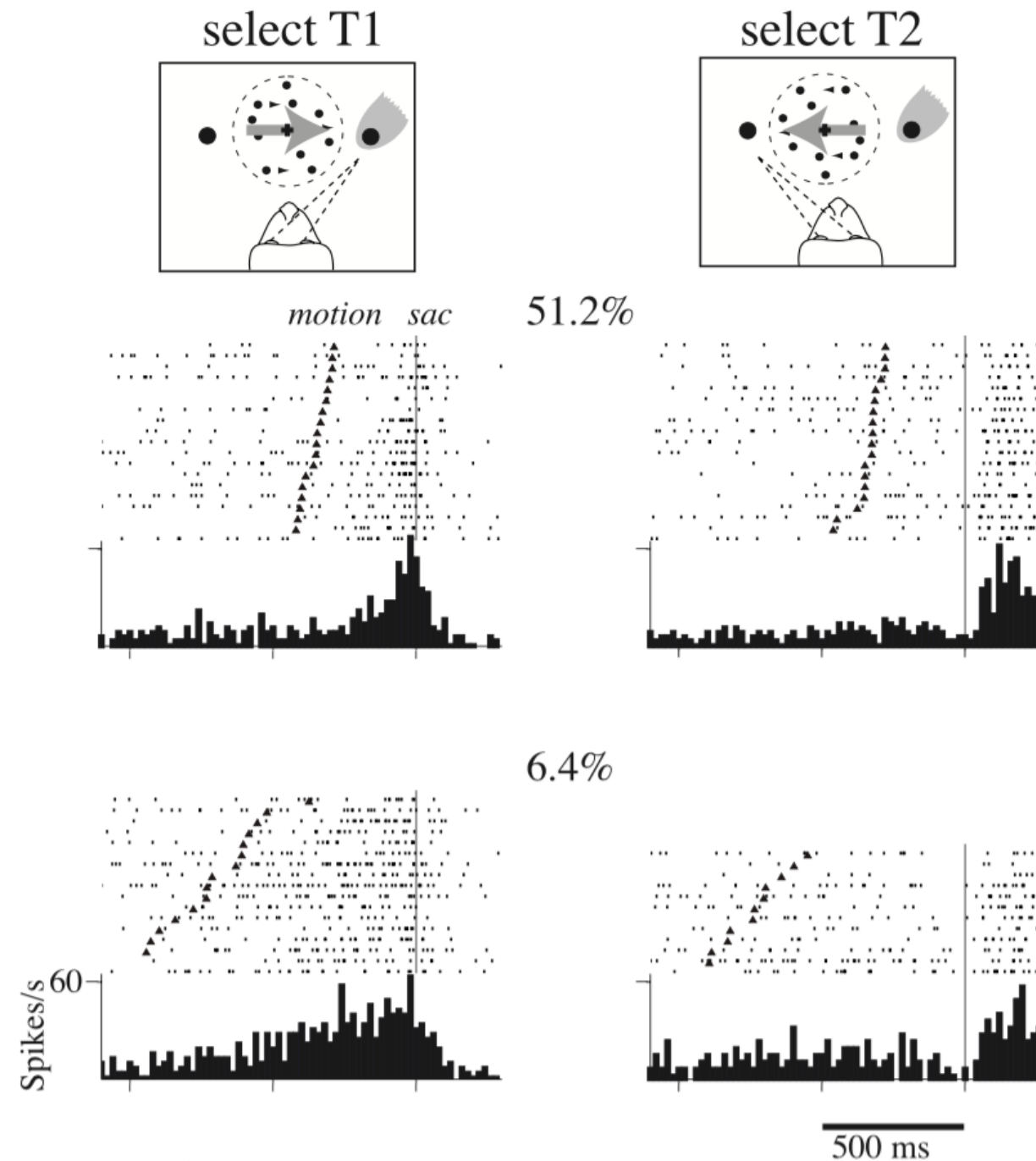
# Computations: example 2

## Perceptual decision-making



# Computations

## Perceptual decision-making



Encoding of decision

[Gold and Shadlen 2007]

# Questions

- How does network structure determine dynamics?
- How do dynamics of neural networks lead to computations?

Connectivity → Dynamics → Computations

**Goal:** build simplified models of working memory  
& decision making

# Method: Dynamical systems

$$\frac{d\mathbf{x}}{dt} = G(\mathbf{x}, \mathbf{u}) \qquad \mathbf{x} = (x_1, \dots, x_N)$$

$\mathbf{u}$  : inputs

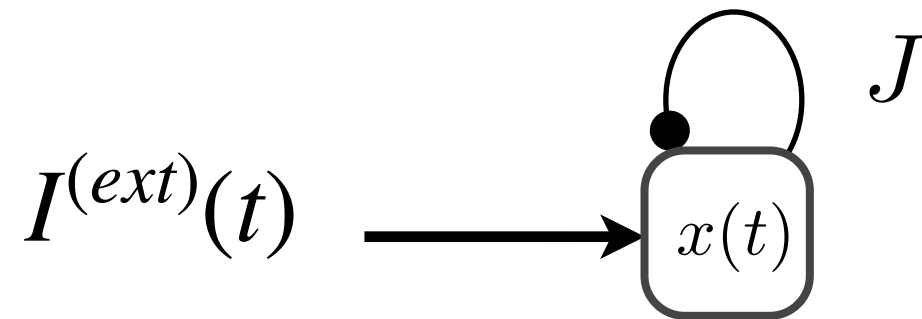
Approach:

1. Determine fixed points
2. Determine stability of fixed points
3. Graphical analysis

# Outline

1. Single Unit
2. Two Units
3. Many Units

# Single unit /population



$$\tau \frac{dx}{dt} = -x + \underline{J \phi(x)} + I^{(ext)}(t)$$

recurrent input = feedback

# Single population - no feedback

$$\tau \frac{dx}{dt} = -x + I^{(ext)}$$

# Single population - no feedback

$$\tau \frac{dx}{dt} = -x + I^{(ext)}$$

I. Fixed point:  $x^*$  such that  $\frac{dx^*}{dt} = 0$

$$\rightarrow x^* = I^{(ext)}$$

# Single population - no feedback

$$\tau \frac{dx}{dt} = -x + I^{(ext)}$$

1. Fixed point:  $x^*$  such that  $\frac{dx^*}{dt} = 0$

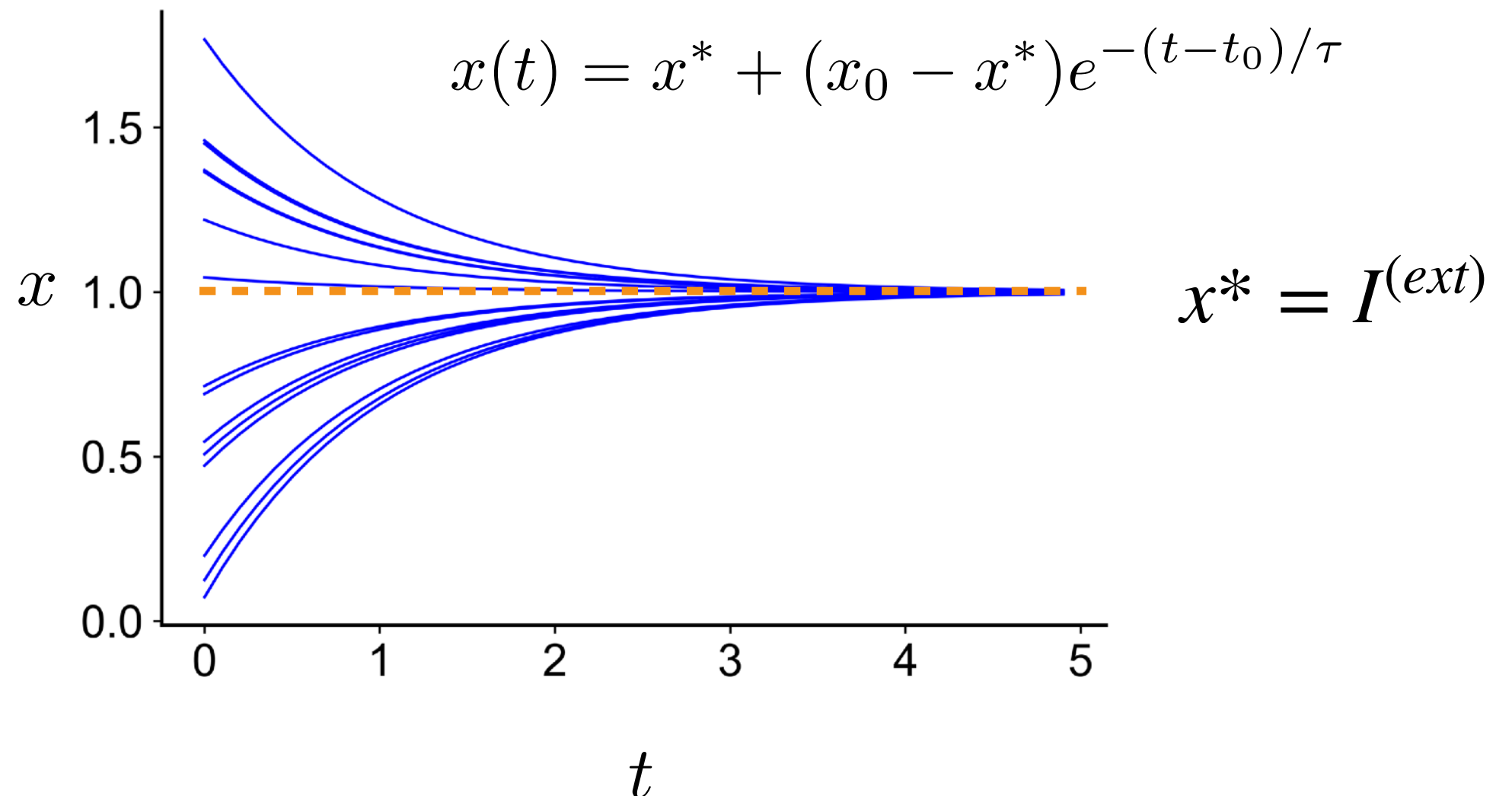
$$\rightarrow x^* = I^{(ext)}$$

2. Full dynamics (starting from  $x_0$  at  $t_0$  )

$$x(t) = x^* + (x_0 - x^*)e^{-(t-t_0)/\tau}$$

# Single population - no feedback

$$\tau \frac{dx}{dt} = -x + I^{(ext)}$$



# Single population, with feedback

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$

# Single population, with feedback, linear

$$\tau \frac{dx}{dt} = -x + Jx + I^{(ext)}$$

# Single population, with feedback, linear

$$\tau \frac{dx}{dt} = -x + Jx + I^{(ext)}$$

I. Fixed point:  $x^*$  such that  $\frac{dx^*}{dt} = 0$

$$x^* = I^{(ext)} / (1 - J)$$

→ Amplification for  $0 < J < 1$

# Single population, with feedback, linear

$$\tau \frac{dx}{dt} = -x + Jx + I^{(ext)}$$

1. Fixed point:  $x^*$  such that  $\frac{dx^*}{dt} = 0$

$$x^* = I^{(ext)} / (1 - J)$$

2. Full dynamics (starting from  $x_0$  at  $t_0$  )

$$x(t) = x^* + (x_0 - x^*)e^{-\frac{(1-J)(t-t_0)}{\tau}}$$

# Single population, with feedback, linear

$$\tau \frac{dx}{dt} = -x + Jx + I^{(ext)}$$

1. Fixed point:  $x^*$  such that  $\frac{dx^*}{dt} = 0$

$$x^* = I^{(ext)} / (1 - J)$$

2. Full dynamics (starting from  $x_0$  at  $t_0$ )

$$x(t) = x^* + (x_0 - x^*) e^{-\frac{(1-J)(t-t_0)}{\tau}}$$



$J < 1$  : stable dynamics, attractive fixed point  $\tau_{eff} = \frac{\tau}{1 - J}$

$J > 1$  : unstable dynamics, repulsive fixed point

# Single population, with feedback, non-linear

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$

I. Fixed point:  $x^* = J \phi(x^*) + I^{(ext)}$

# Single population, with feedback, non-linear

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$

1. Fixed point:  $x^* = J \phi(x^*) + I^{(ext)}$

2. Linear stability  $x(t) = x^* + x^{(1)}(t)$

# Single population, with feedback, non-linear

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$

1. Fixed point:  $x^* = J \phi(x^*) + I^{(ext)}$

2. Linear stability  $x(t) = x^* + x^{(1)}(t)$

$$\tau \frac{dx^{(1)}}{dt} = -x^{(1)} + \underbrace{J \phi'(x^*)}_{\text{effective coupling}} x^{(1)}$$

**effective coupling**  $J_{eff} = J \phi'(x^*)$

# Single population, with feedback, non-linear

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$

1. Fixed point:  $x^* = J \phi(x^*) + I^{(ext)}$

2. Linear stability  $x(t) = x^* + x^{(1)}(t)$

$$\tau \frac{dx^{(1)}}{dt} = -x^{(1)} + \underbrace{J \phi'(x^*)}_{\text{effective coupling}} x^{(1)}$$

**effective coupling**  $J_{eff} = J \phi'(x^*)$

$J_{eff} < 1$  : stable, attractive fixed point  $\tau_{eff} = \frac{\tau}{1 - J_{eff}}$

$J_{eff} > 1$  : unstable, repulsive fixed point

# Dynamical systems

$$\frac{d\mathbf{x}}{dt} = G(\mathbf{x}, \mathbf{u}) \quad \mathbf{x} = (x_1, \dots, x_P)$$

$\mathbf{u}$  : inputs

Approach:

1. Determine fixed points
2. Determine stability of fixed points
3. Graphical analysis

Local Analyses

Global

# Single population, graphical analysis

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$

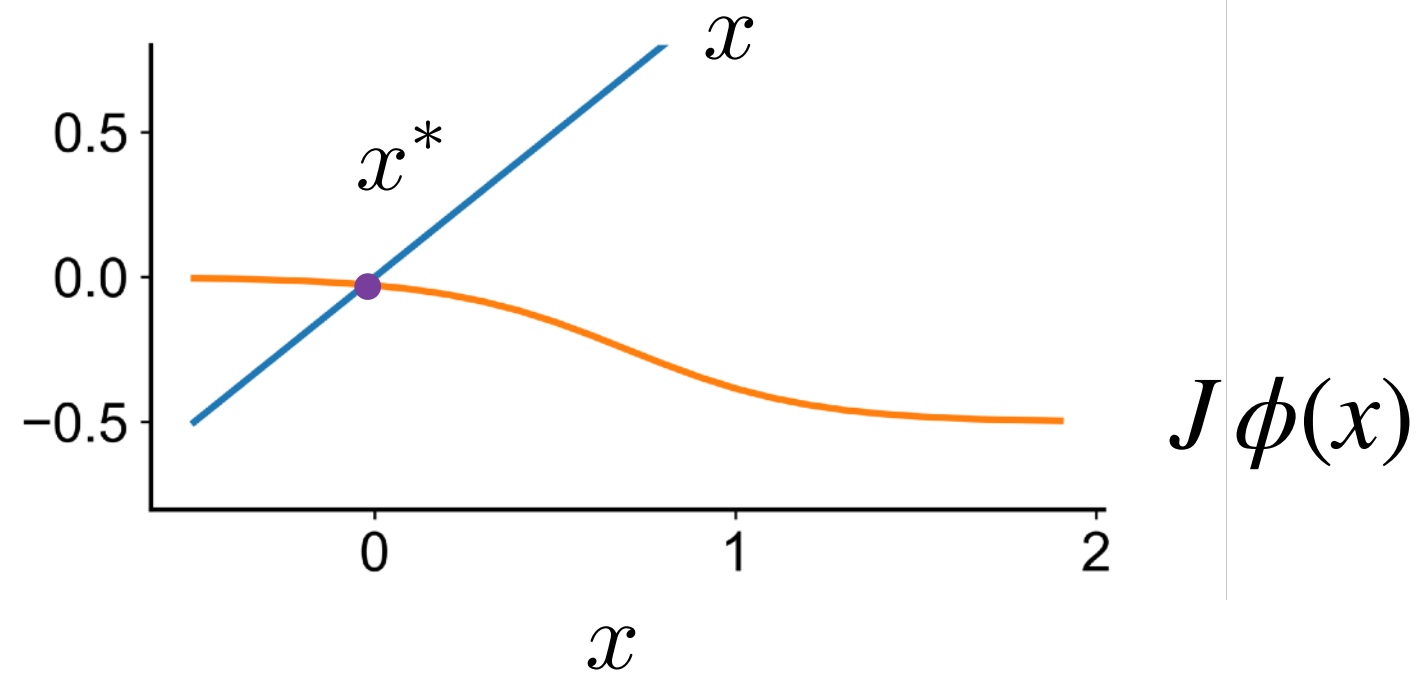
Approach 1:  $x = J \phi(x) + I^{(ext)}$

Approach 2:  $\frac{dx}{dt}$

# Single population, inhibitory

$$J < 0$$

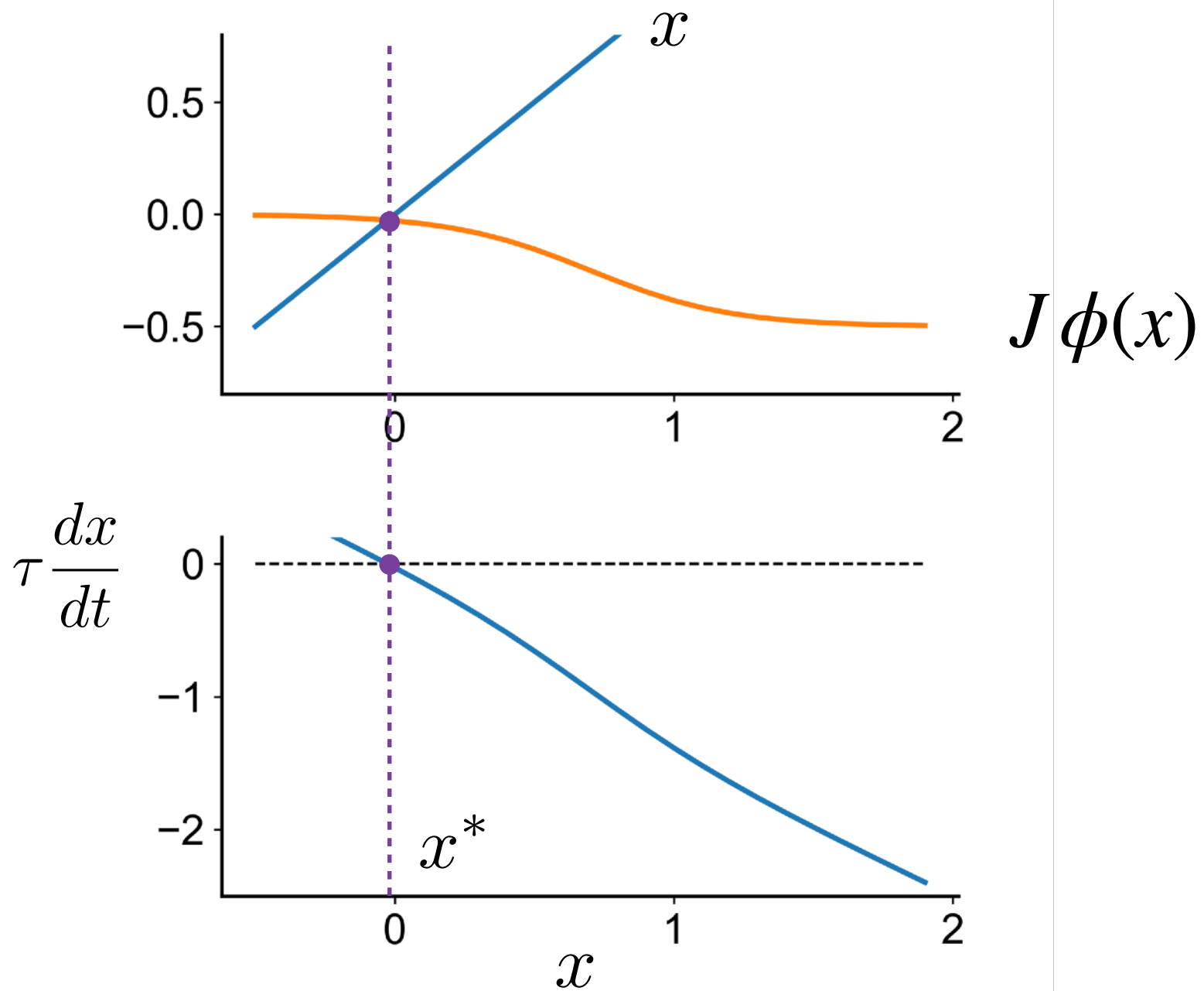
$$\tau \frac{dx}{dt} = -x + J \phi(x)$$



# Single population, inhibitory

$$J < 0$$

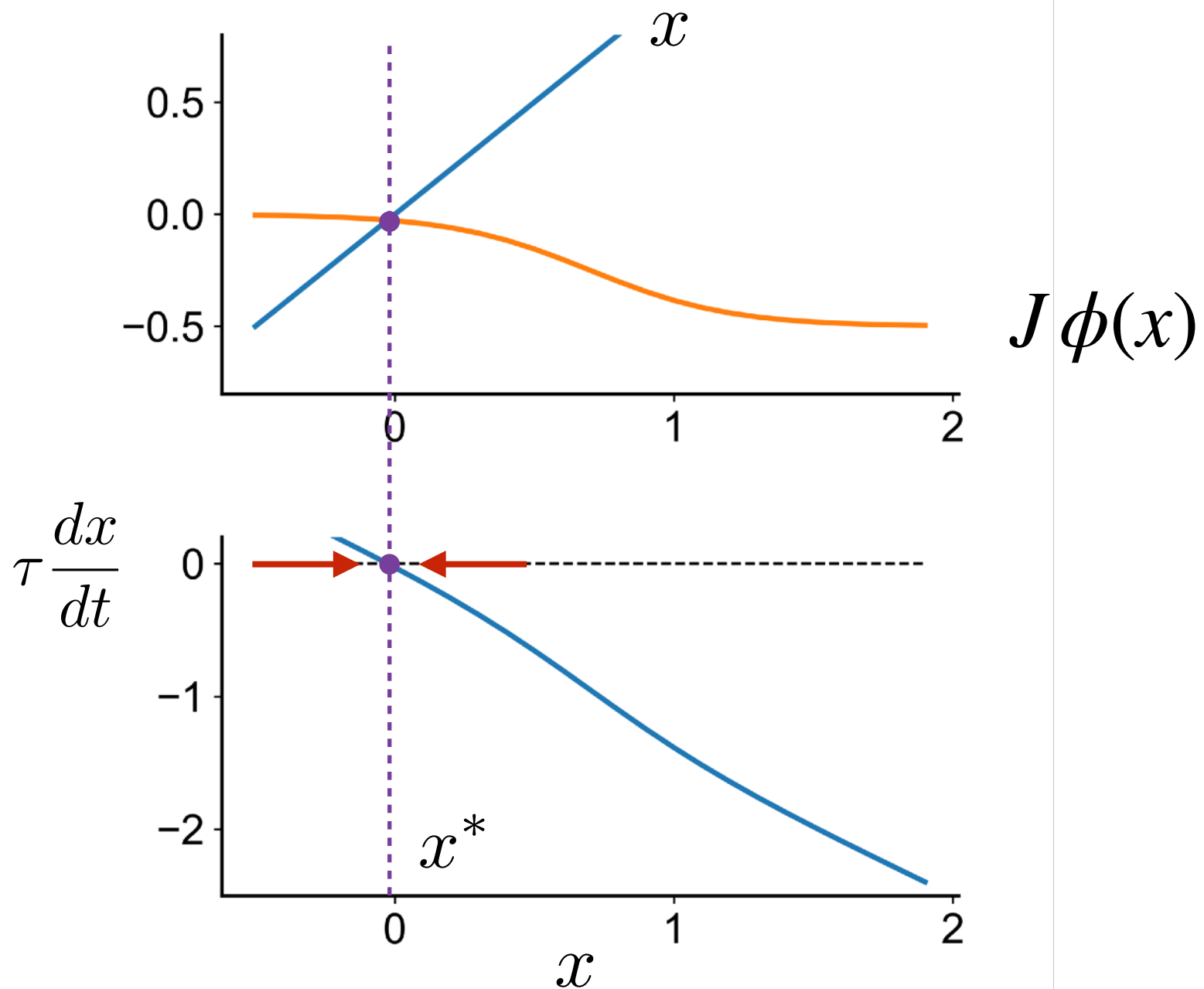
$$\tau \frac{dx}{dt} = -x + J \phi(x)$$



# Single population, inhibitory

$$J < 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x)$$

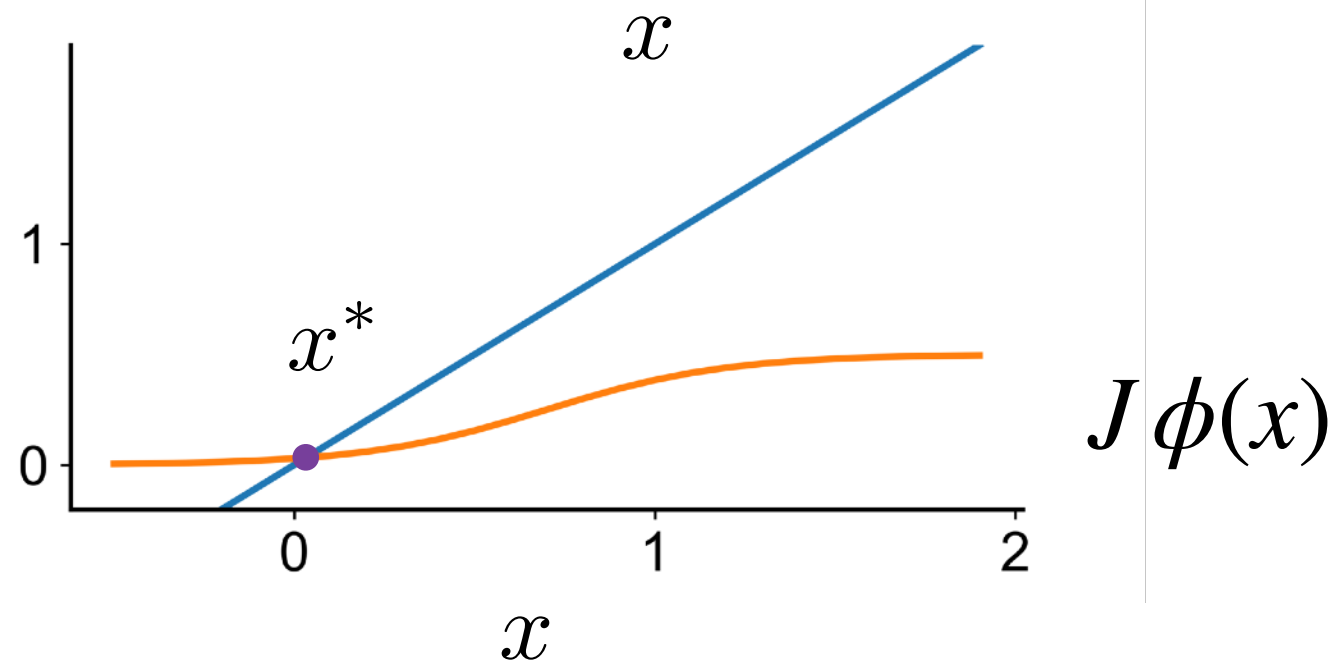


Single fixed point, always stable

# Single population, excitatory

$$J > 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x)$$

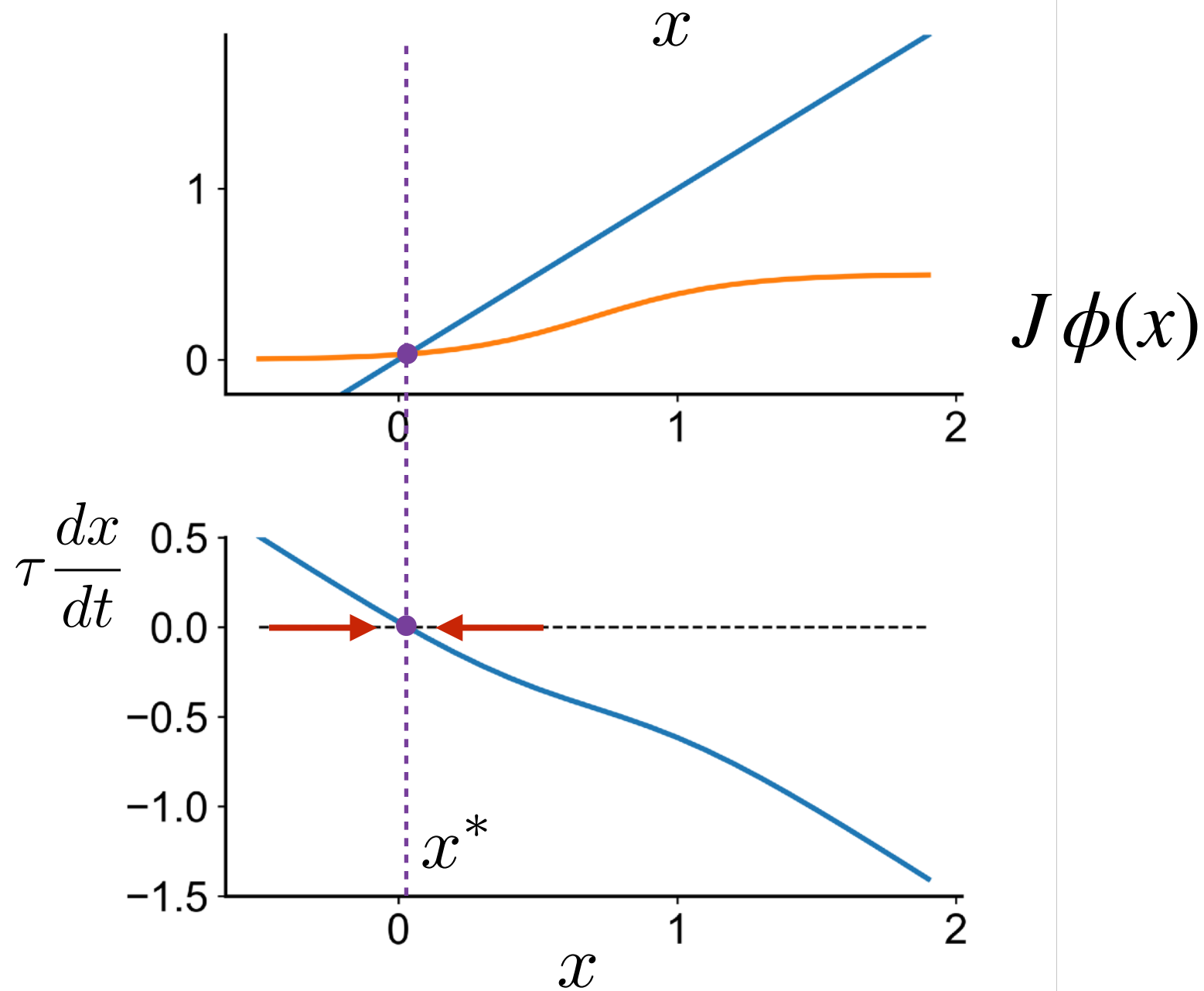


Single fixed point

# Single population, excitatory

$$J > 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x)$$

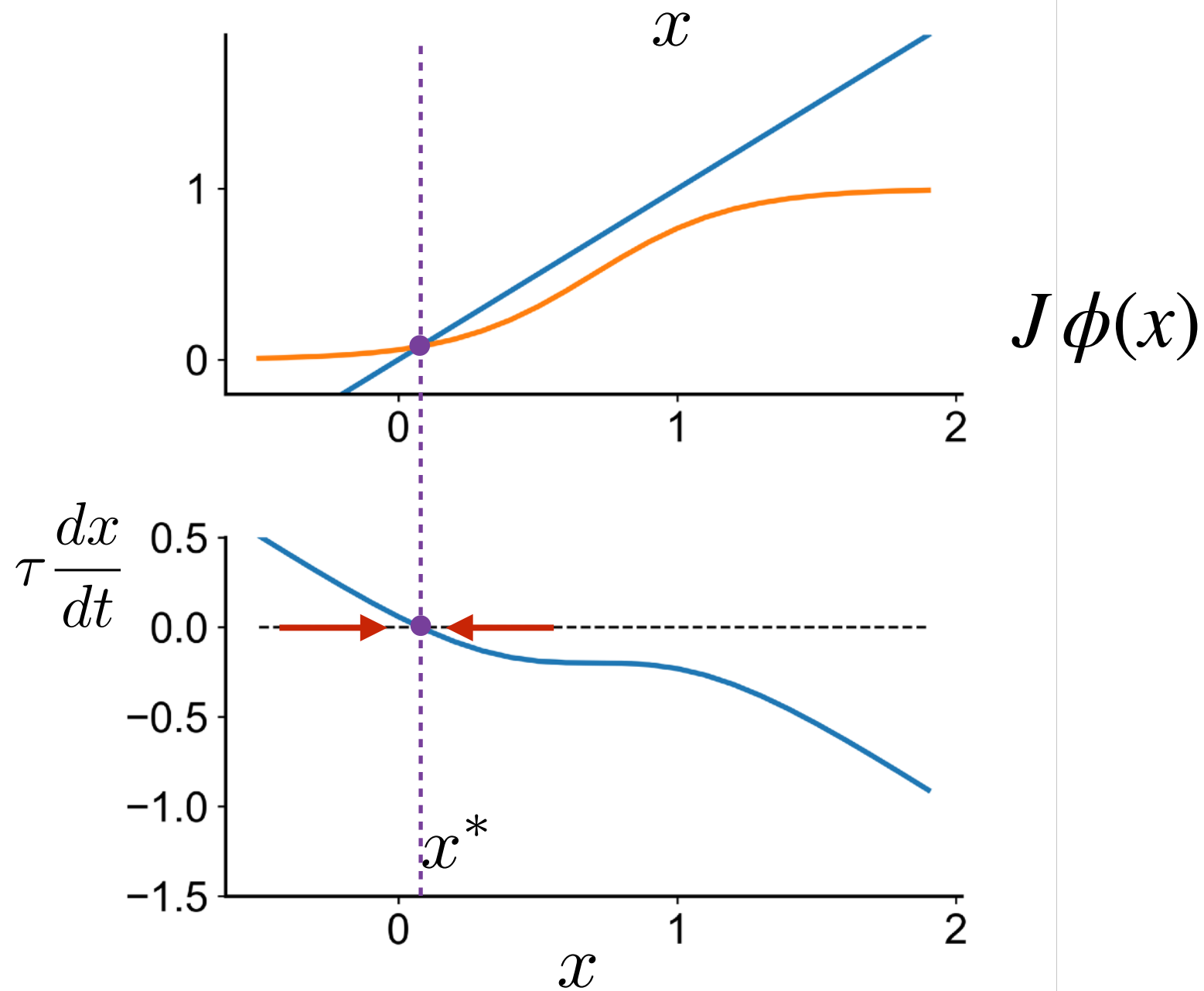


Single fixed point

# Single population, excitatory

$$J > 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x)$$

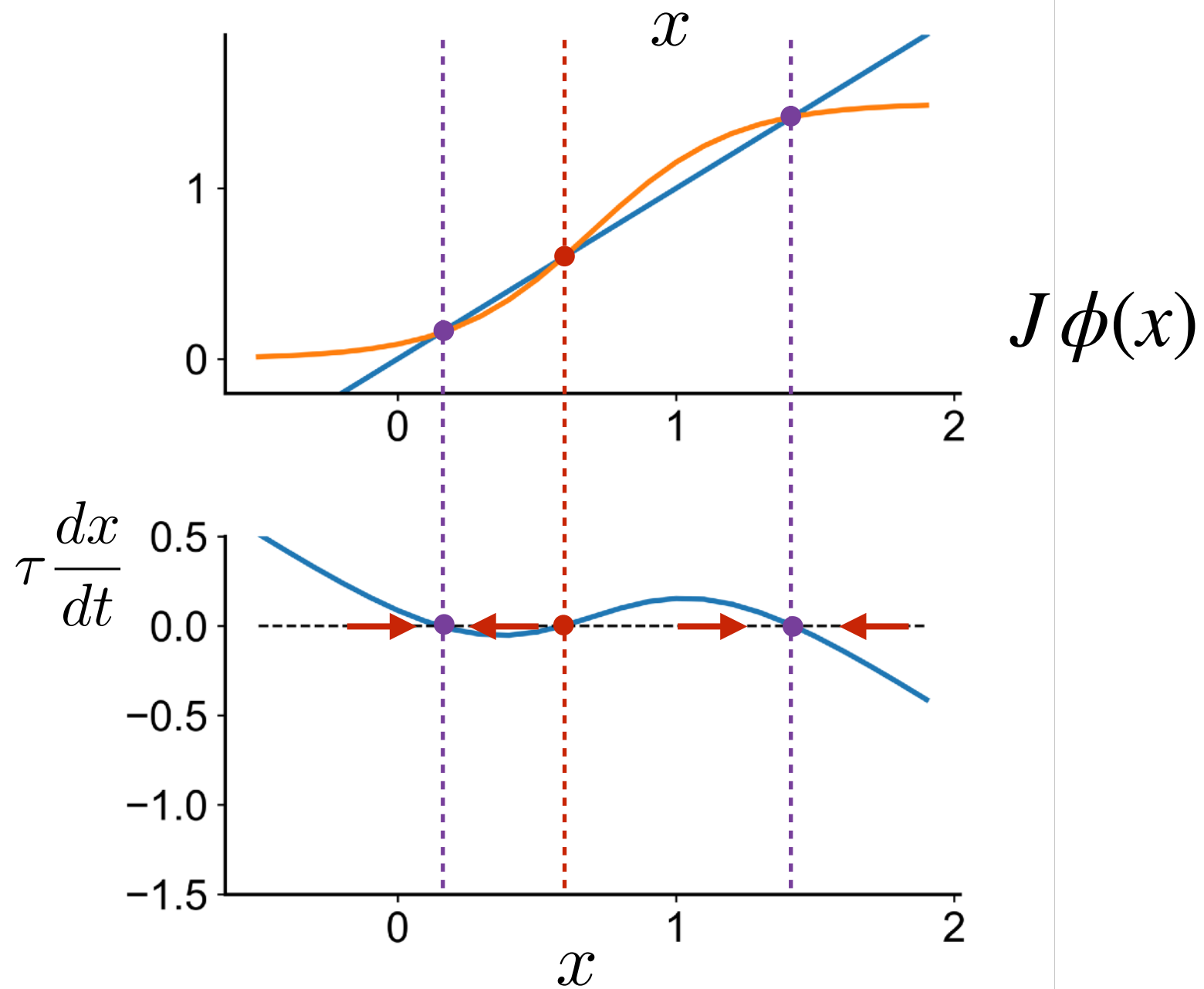


Single fixed point

# Single population, excitatory

$$J > 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x)$$

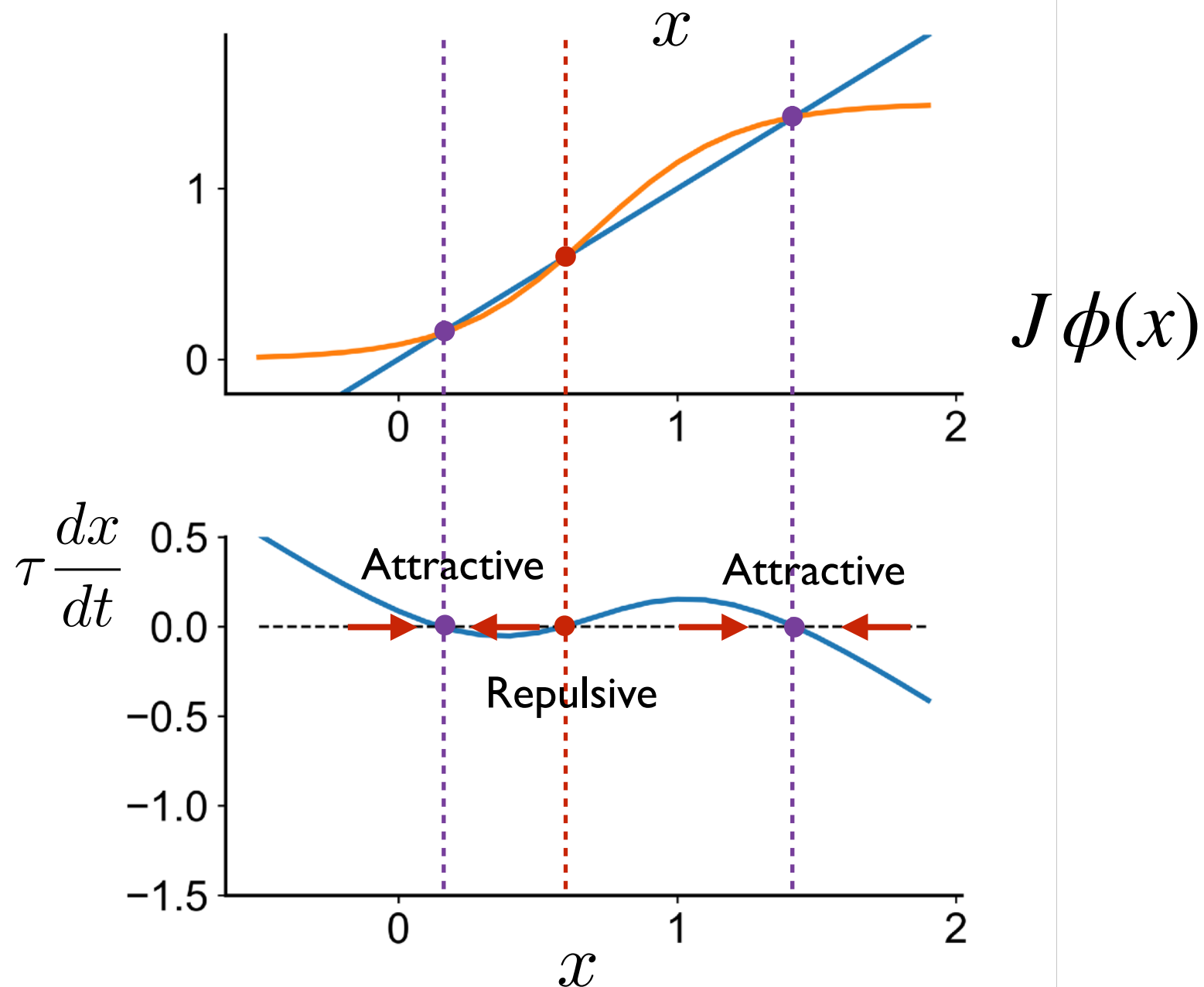


Three fixed points

# Single population, excitatory

$$J > 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x)$$

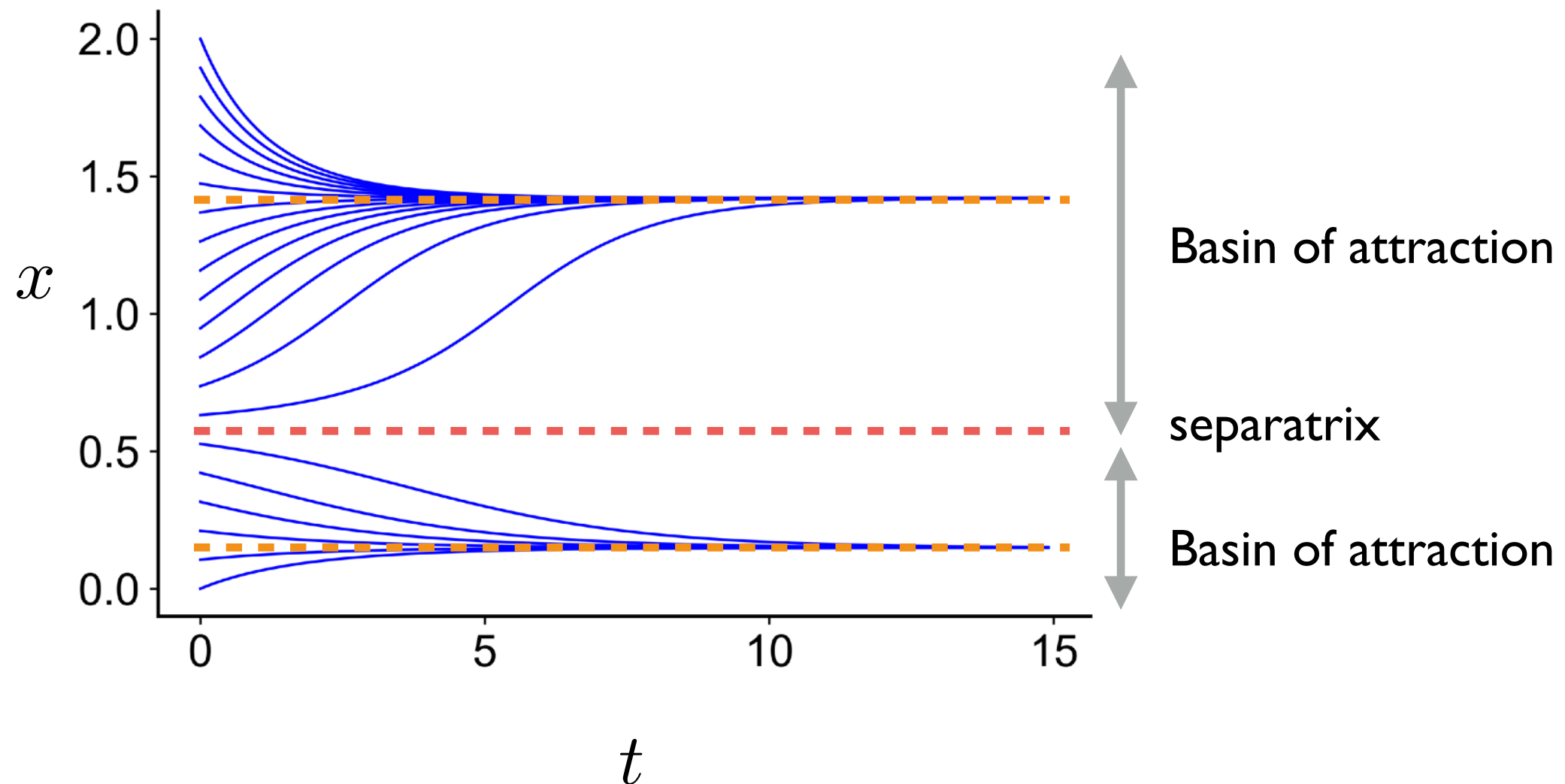


Three fixed points, 2 attractors

# Multi-stability

$$J > 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x)$$



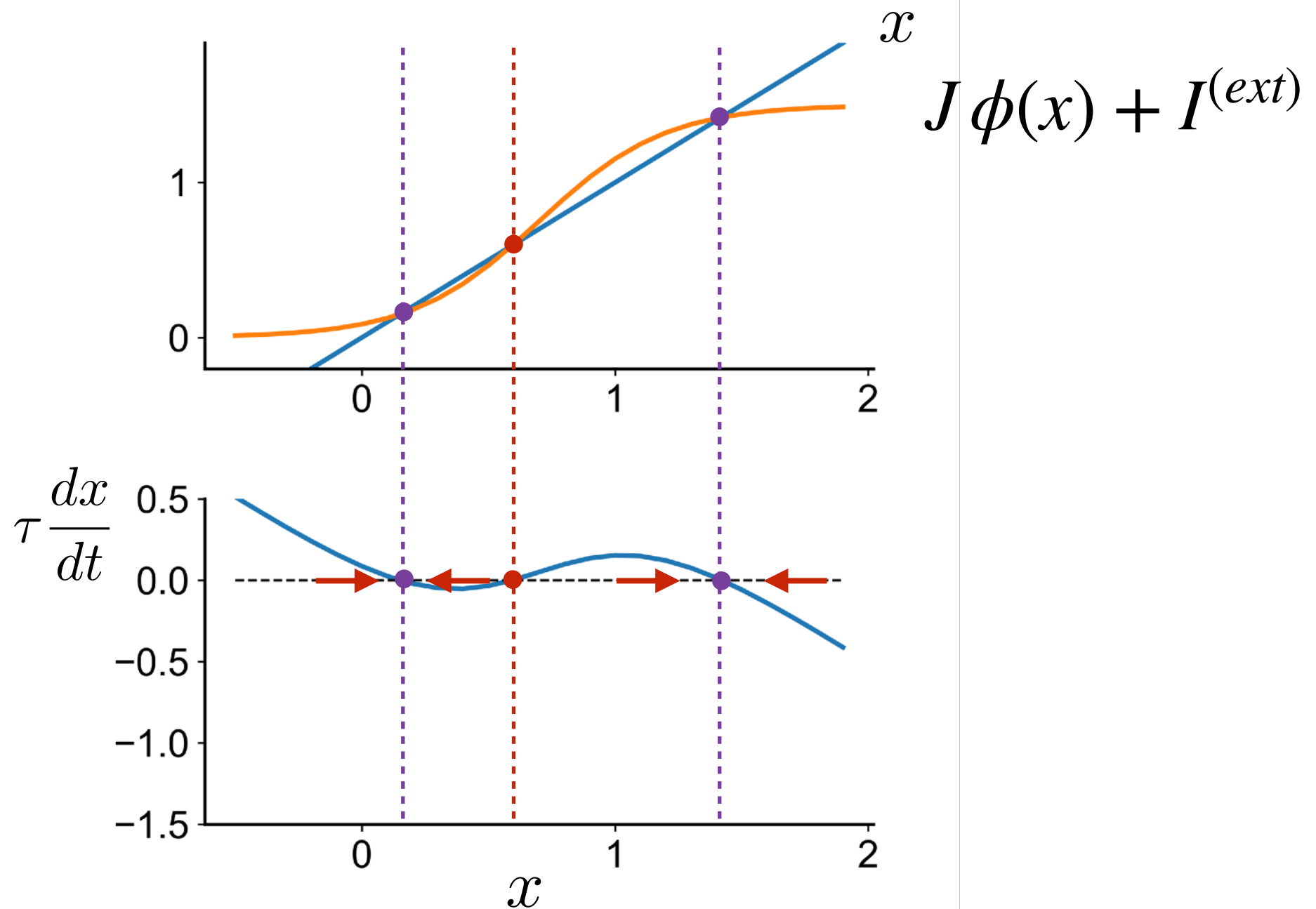
Final state depends on initial condition: memory

# Effects of inputs

$$J > 0$$

$$\mu^{ext} = 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$



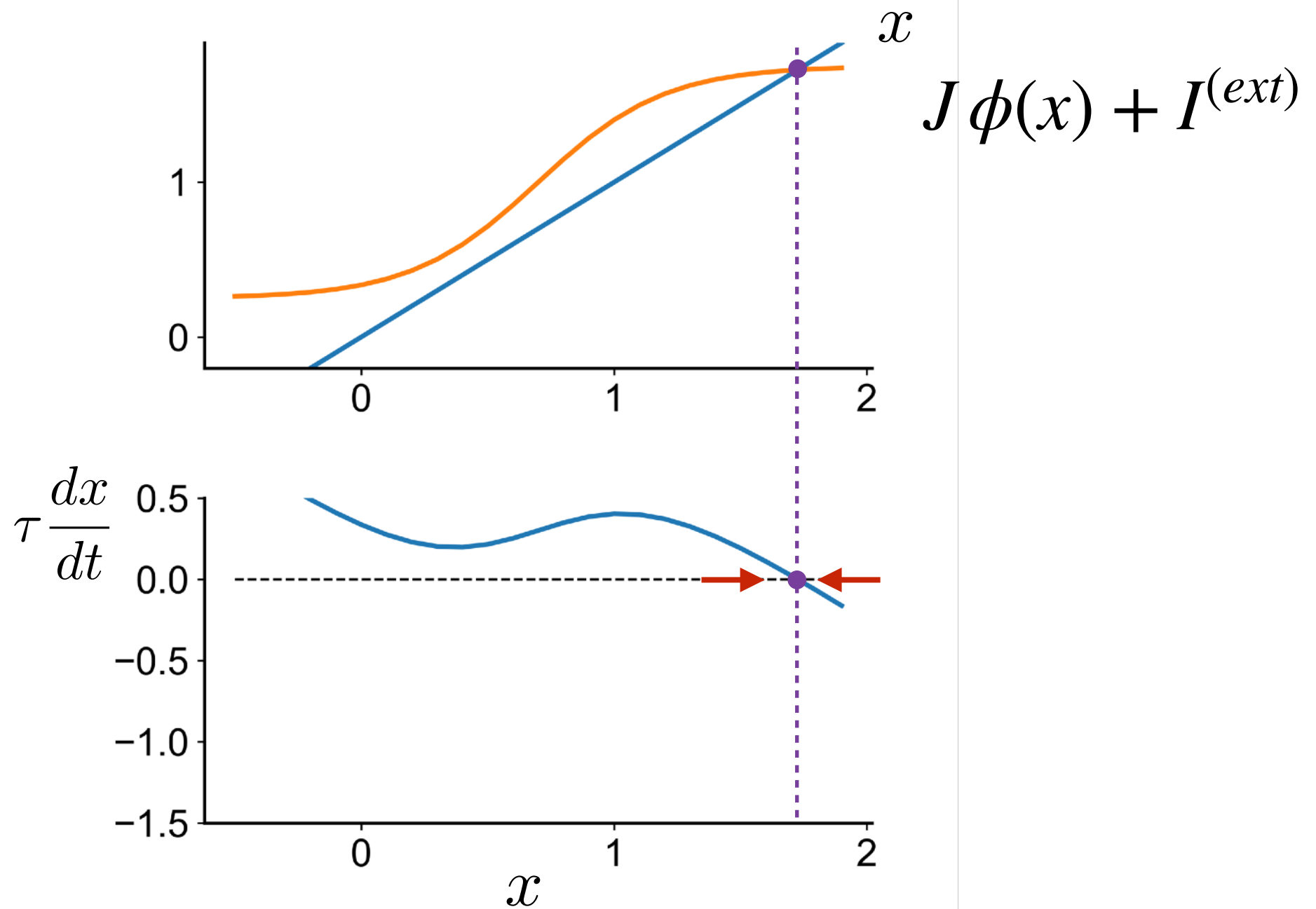
Bi-stability

# Effects of inputs

$$J > 0$$

$$\mu^{ext} > 0$$

$$\tau \frac{dx}{dt} = -x + J\phi(x) + I^{(ext)}$$

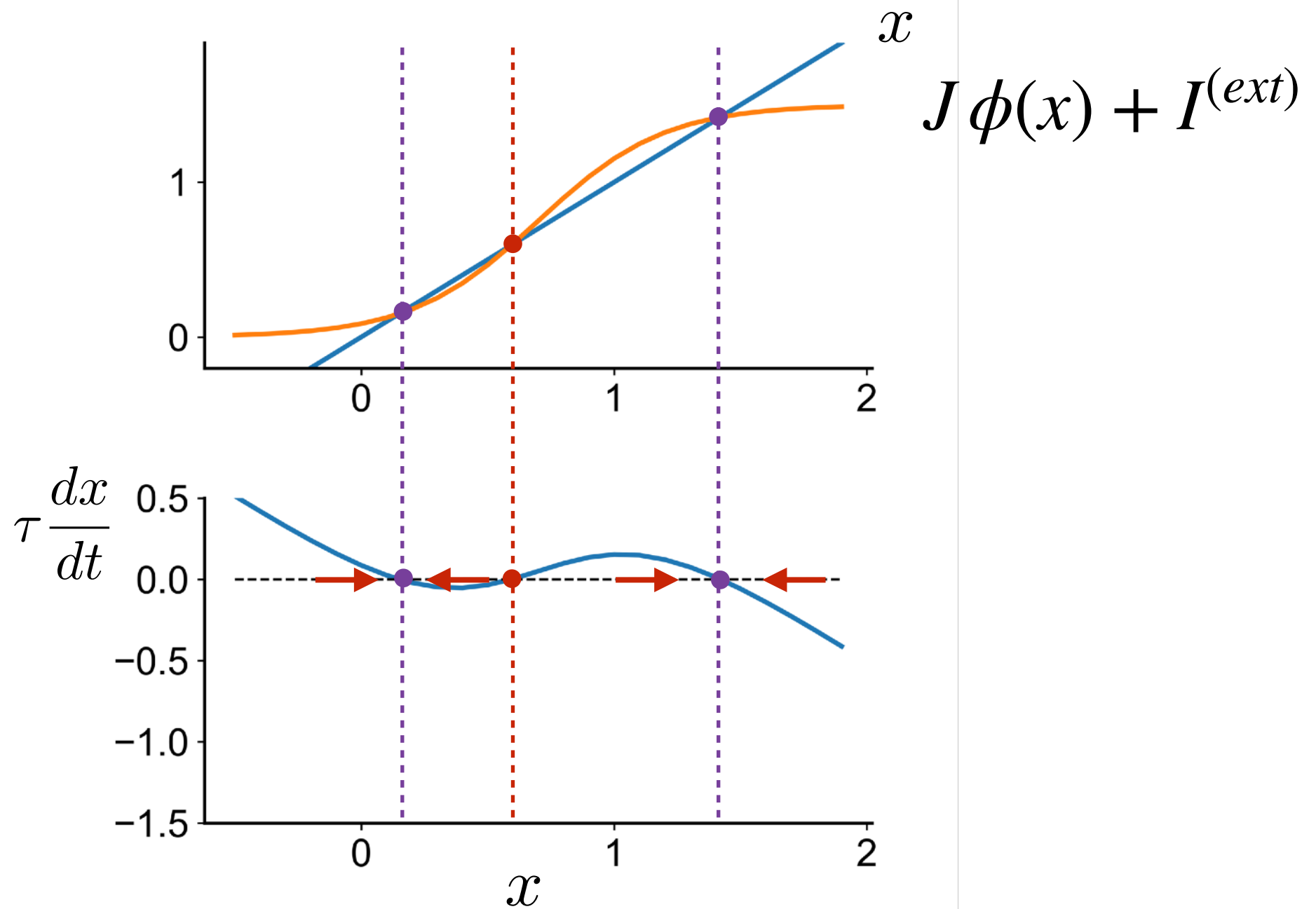


# Effects of inputs

$$J > 0$$

$$\mu^{ext} = 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$



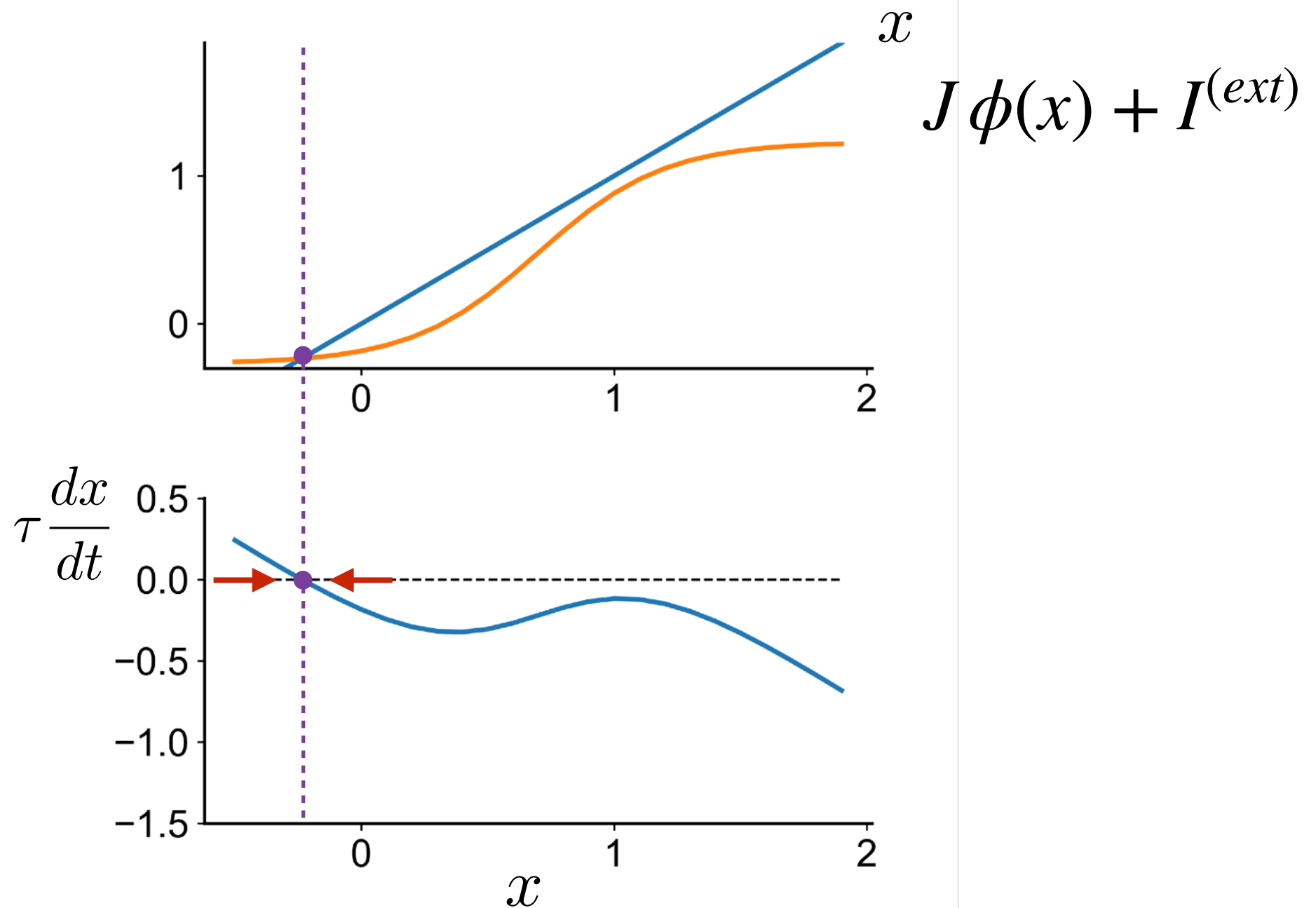
Bi-stability

# Effects of inputs

$$J > 0$$

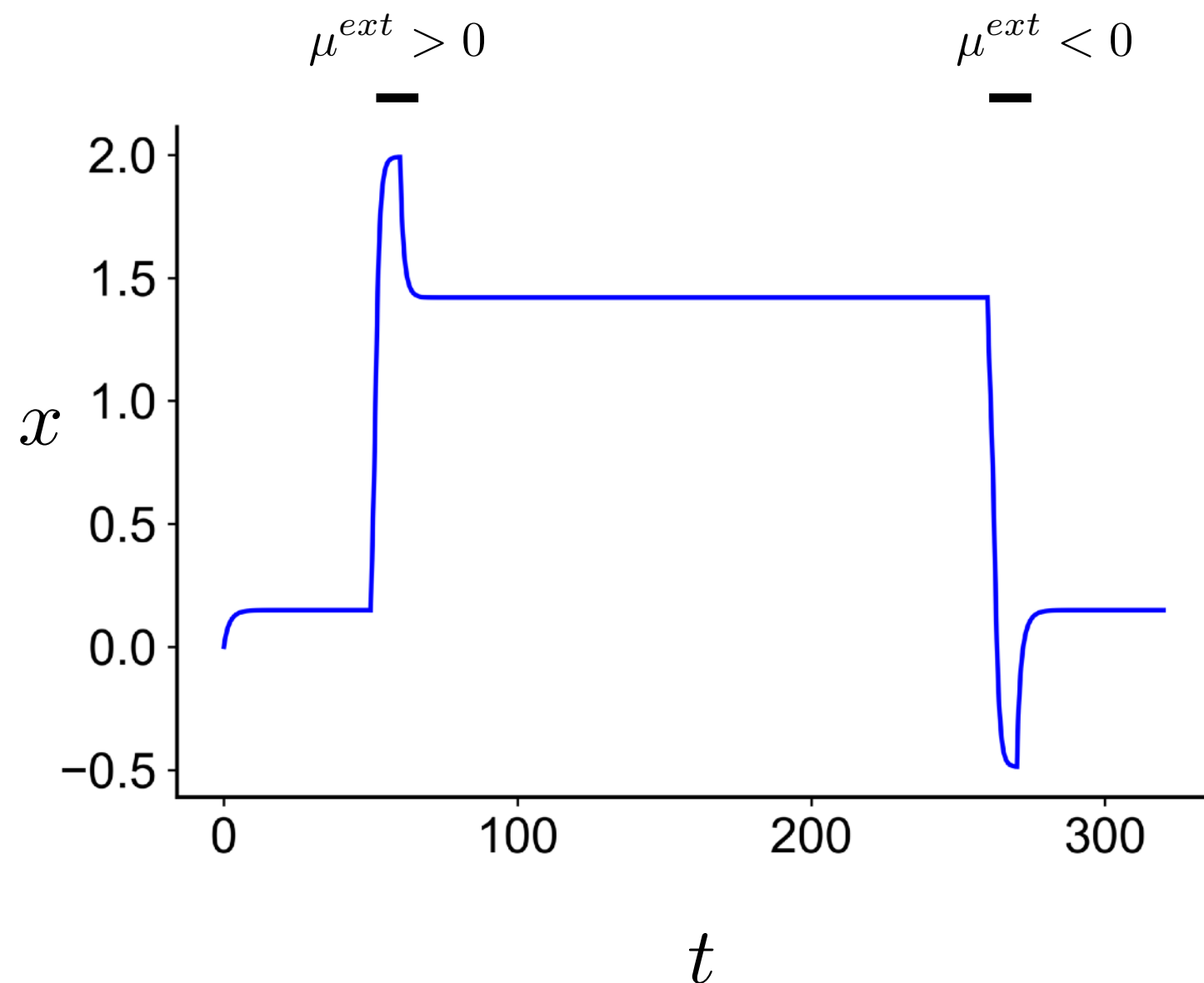
$$\mu^{ext} < 0$$

$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$



# Memory

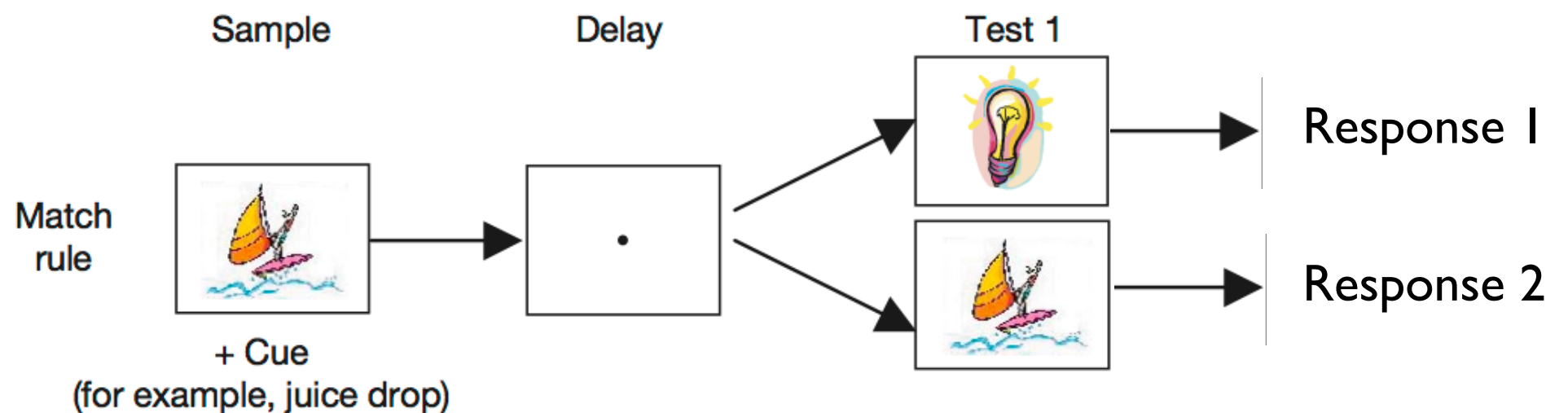
$$\tau \frac{dx}{dt} = -x + J \phi(x) + I^{(ext)}$$



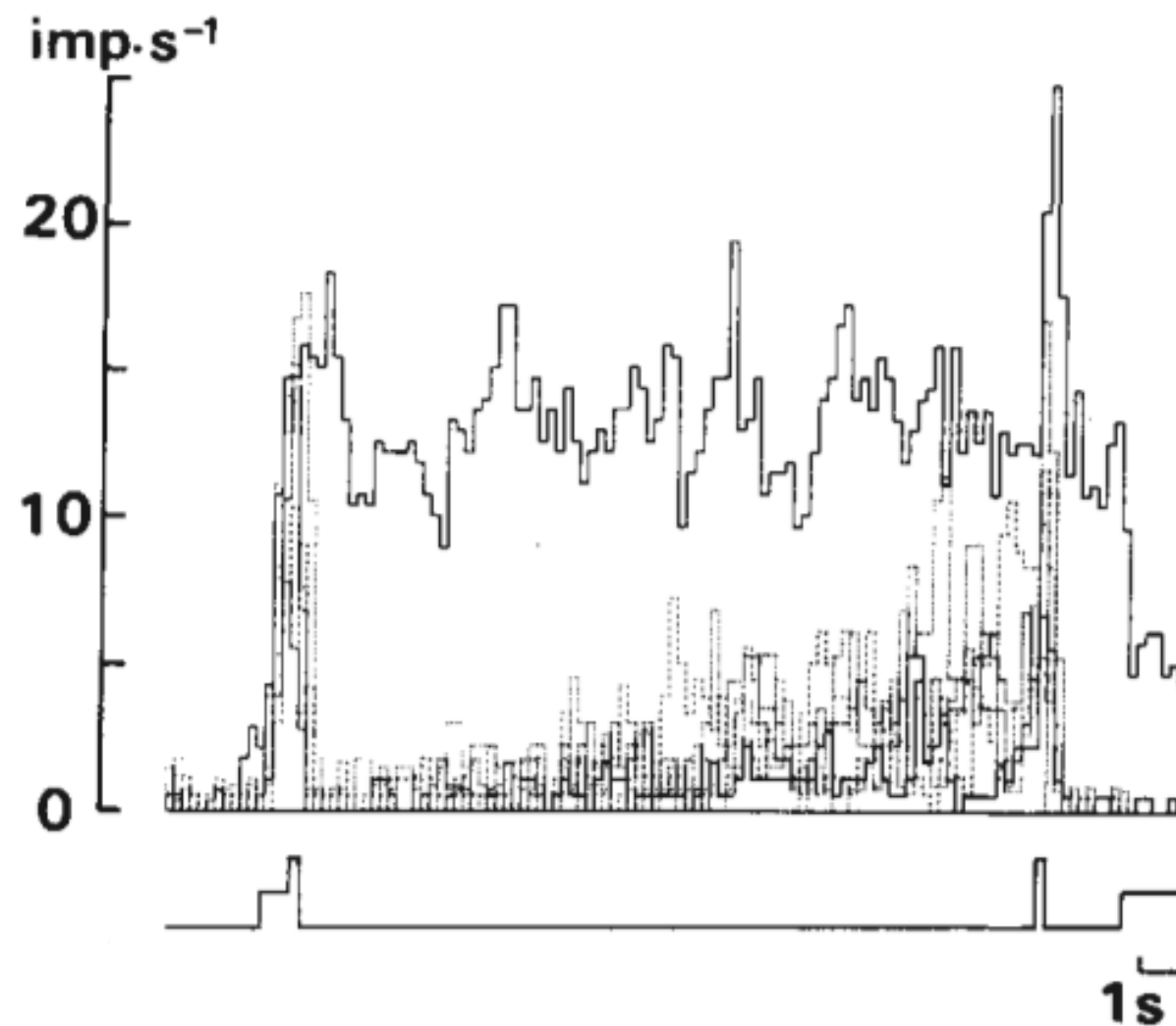
Memory from positive feedback

# From neural activity to behaviour

## Delayed match-to sample task



# Persistent mnemonic activity



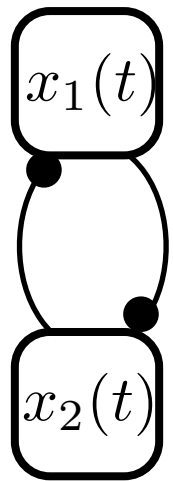
➔ Working memory encoded in persistent activity

[Miyashita and Chang 1988]

# Outline

1. Single Unit
- 2. Two Units**
3. Many Units

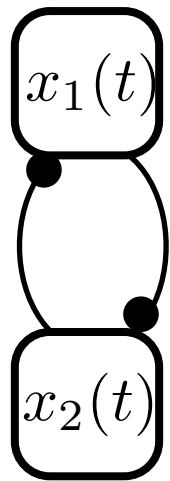
# Two populations



$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2) + I_1^{(ext)}$$

$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1) + I_2^{(ext)}$$

# Two populations



$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2) + I_1^{(ext)}$$

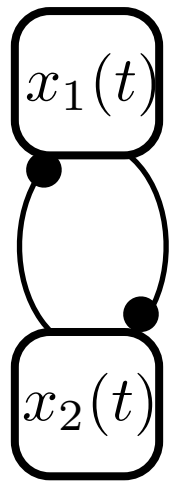
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1) + I_2^{(ext)}$$

I. Fixed point:  $(x_1^*, x_2^*)$  such that  $\frac{dx_1^*}{dt} = 0$  and  $\frac{dx_2^*}{dt} = 0$

$$x_1^* = J \phi(x_2^*) + I_1^{(ext)}$$

$$x_2^* = J \phi(x_1^*) + I_2^{(ext)}$$

# Two populations



$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2) + I_1^{(ext)}$$
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1) + I_2^{(ext)}$$

1. Fixed point:  $(x_1^*, x_2^*)$  such that  $\frac{dx_1^*}{dt} = 0$  and  $\frac{dx_2^*}{dt} = 0$

$$x_1^* = J \phi(x_2^*) + I_1^{(ext)}$$

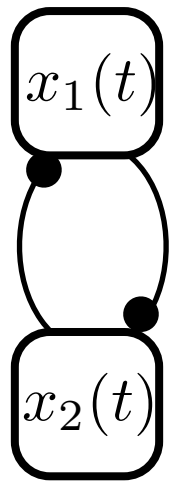
$$x_2^* = J \phi(x_1^*) + I_2^{(ext)}$$

2. Linear stability

$$x_1(t) = x_1^* + x_1^{(1)}(t)$$

$$x_2(t) = x_2^* + x_2^{(1)}(t)$$

# Two populations



$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2) + I_1^{(ext)}$$

$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1) + I_2^{(ext)}$$

1. Fixed point:  $(x_1^*, x_2^*)$  such that  $\frac{dx_1^*}{dt} = 0$  and  $\frac{dx_2^*}{dt} = 0$

$$x_1^* = J \phi(x_2^*) + I_1^{(ext)}$$

$$x_2^* = J \phi(x_1^*) + I_2^{(ext)}$$

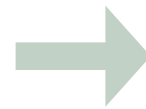
2. Linear stability  $\mathbf{x}^{(1)} = (x_1^{(1)}, x_2^{(1)})$

$$\frac{d\mathbf{x}^{(1)}}{dt} = \mathbf{A} \mathbf{x}^{(1)}$$

→ Stability given by eigenvalues of  $\mathbf{A}$  (“jacobian”)

# Linear analysis

$$\begin{aligned}\tau \frac{dx_1^{(1)}}{dt} &= -x_1^{(1)} + J \phi'(x_2^*) x_2^{(1)} \\ \tau \frac{dx_2^{(1)}}{dt} &= -x_2^{(1)} + J \phi'(x_1^*) x_1^{(1)}\end{aligned}$$



$$\frac{d\mathbf{x}^{(1)}}{dt} = \overbrace{\begin{bmatrix} -1 & JF'(x_2^*) \\ JF'(x_1^*) & -1 \end{bmatrix}}^{\mathbf{A}} \mathbf{x}^{(1)}$$



eigenvalues

$\lambda_1, \lambda_2$

eigenvectors

$\mathbf{u}^{(1)}, \mathbf{u}^{(2)}$

# Linear analysis

$$\begin{aligned}\tau \frac{dx_1^{(1)}}{dt} &= -x_1^{(1)} + J\phi'(x_2^*)x_2^{(1)} \\ \tau \frac{dx_2^{(1)}}{dt} &= -x_2^{(1)} + J\phi'(x_1^*)x_1^{(1)}\end{aligned} \quad \Rightarrow \quad \frac{d\mathbf{x}^{(1)}}{dt} = \overbrace{\begin{bmatrix} -1 & JF'(x_2^*) \\ JF'(x_1^*) & -1 \end{bmatrix}}^{\mathbf{A}} \mathbf{x}^{(1)}$$

eigenvalues  $\lambda_1, \lambda_2$

eigenvectors  $\mathbf{u}^{(1)}, \mathbf{u}^{(2)}$

$$\mathbf{x}^{(1)}(\mathbf{t}) = \alpha_1(t)\mathbf{u}^{(1)} + \alpha_2(t)\mathbf{u}^{(2)}$$

$$\tau \frac{d\alpha_1}{dt} = \lambda_1 \alpha_1$$

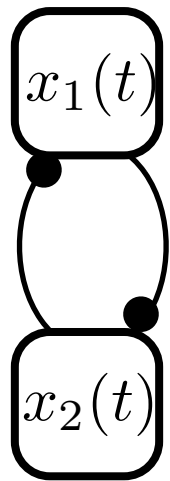
$$\tau \frac{d\alpha_2}{dt} = \lambda_2 \alpha_2$$

$Re(\lambda_1) < 0$      $Re(\lambda_2) < 0$     **stable**

$Re(\lambda_1) > 0$      $Re(\lambda_2) < 0$     **saddle**

$Re(\lambda_1) > 0$      $Re(\lambda_2) > 0$     **unstable**

# Two populations - graphical analysis



$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2) + I_1^{(ext)}$$
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1) + I_2^{(ext)}$$

I. Fixed point:  $(x_1^*, x_2^*)$  such that  $\frac{dx_1^*}{dt} = 0$  and  $\frac{dx_2^*}{dt} = 0$

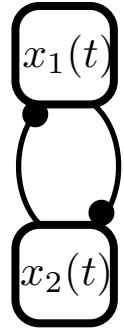
$$x_1^* = J \phi(x_2^*) + I_1^{(ext)}$$

$$x_2^* = J \phi(x_1^*) + I_2^{(ext)}$$

→ nullclines

# Two excitatory populations

$$J > 0$$



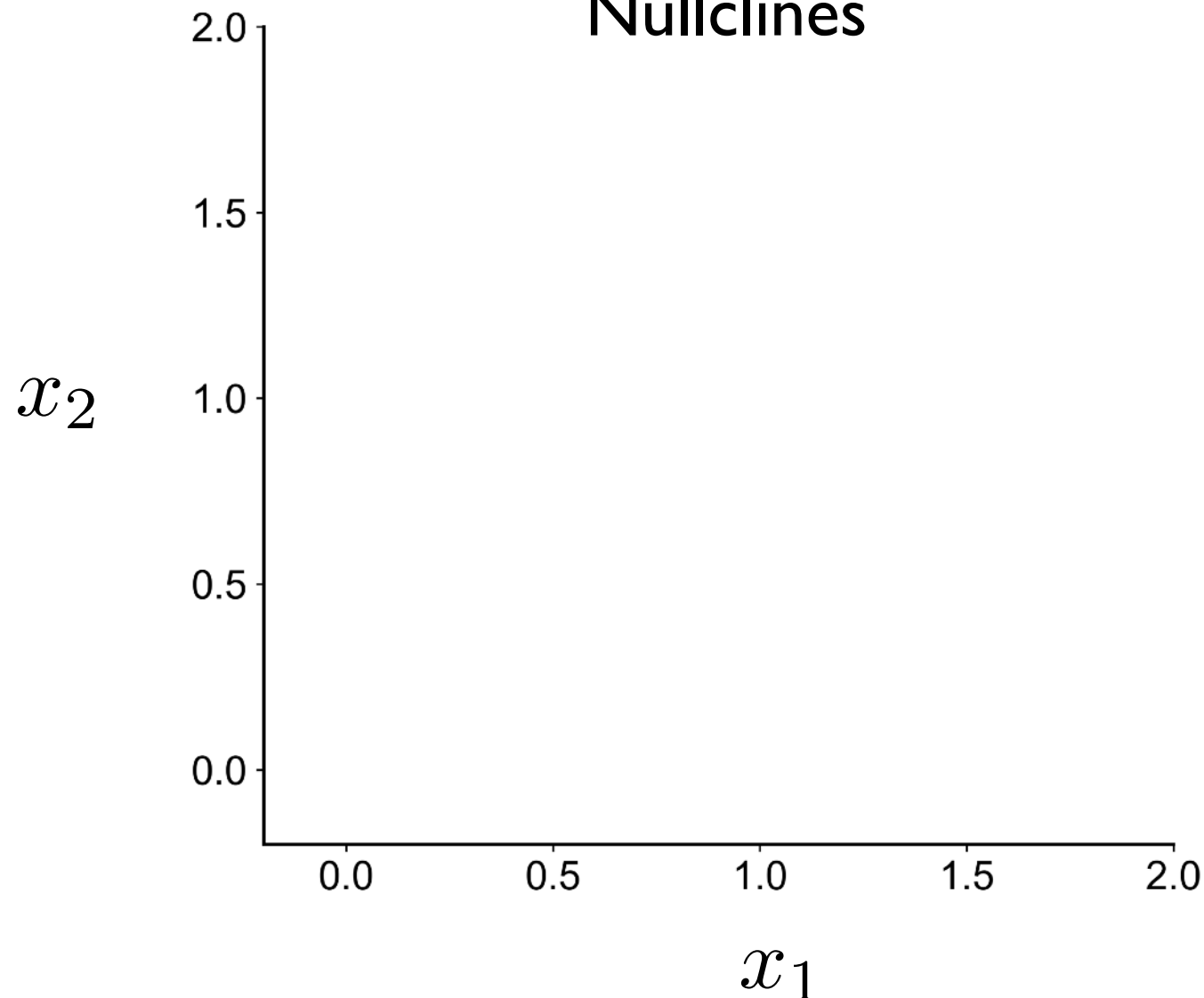
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$

$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$



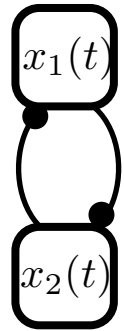
$$x_2 = J \phi(x_1)$$

Nullclines



# Two excitatory populations

$$J > 0$$



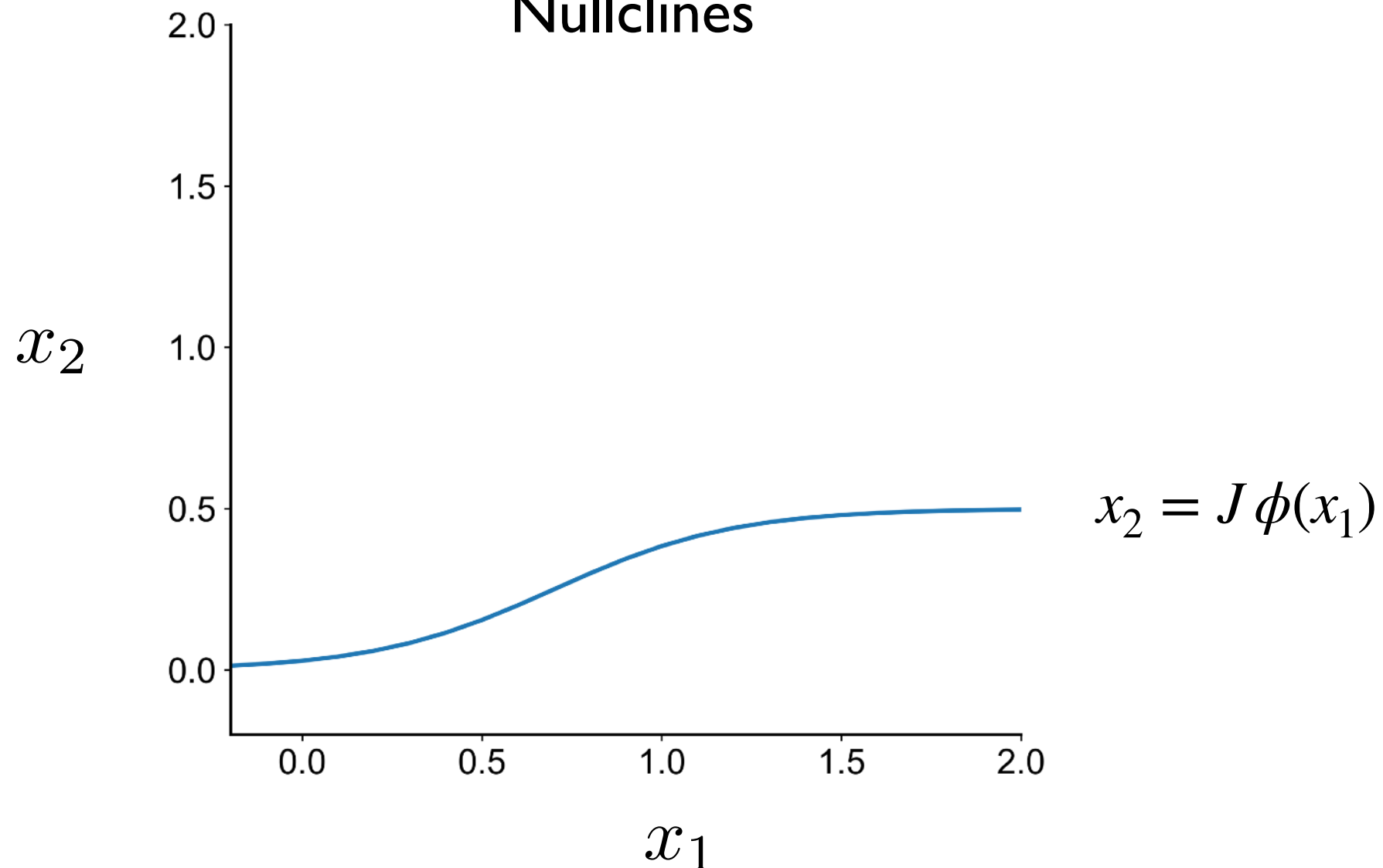
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$



$$x_1 = J \phi(x_2)$$

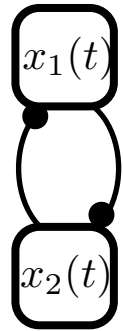
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$

Nullclines



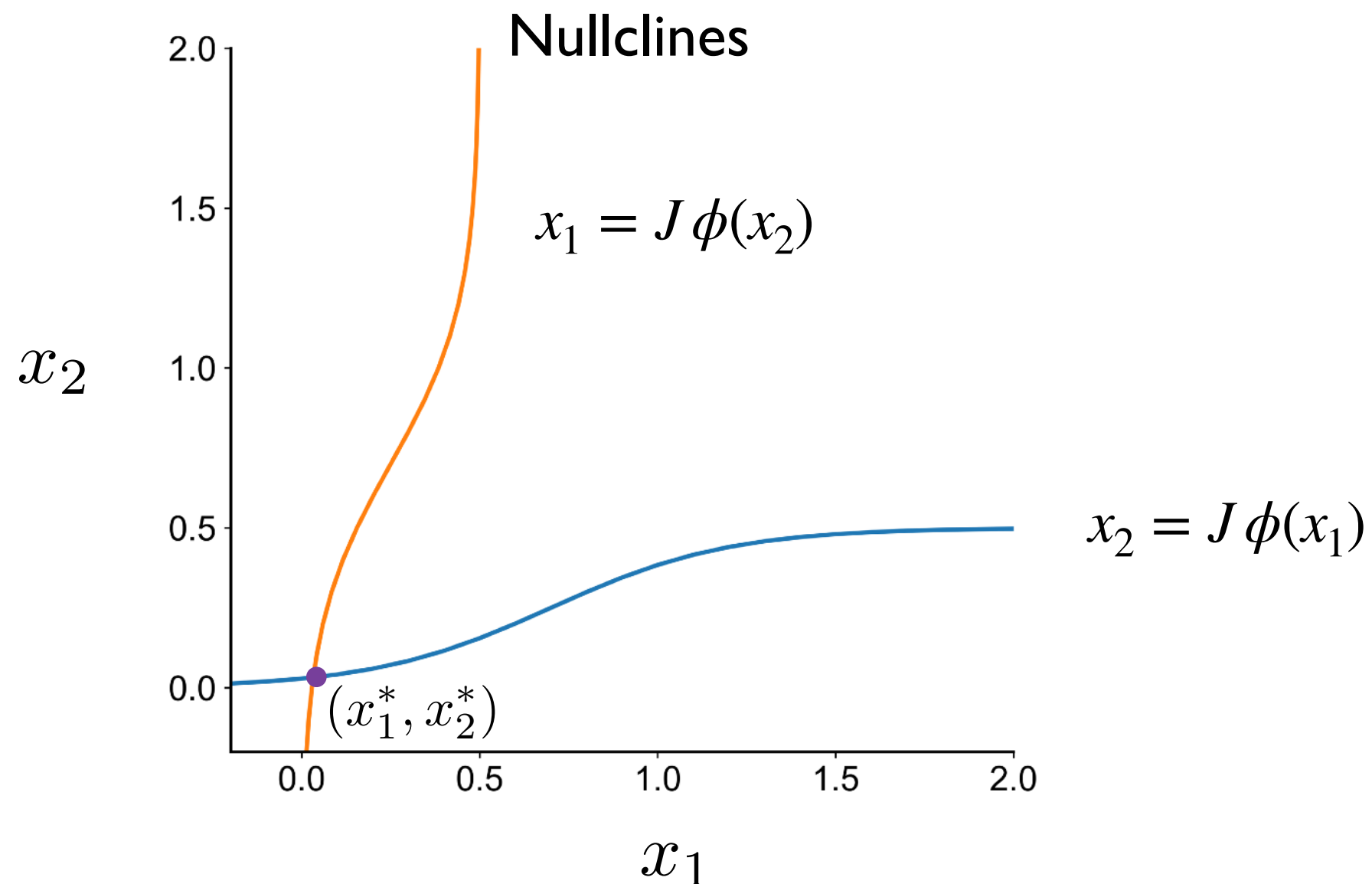
# Two excitatory populations

$$J > 0$$



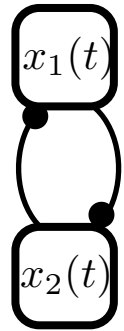
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$

$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$



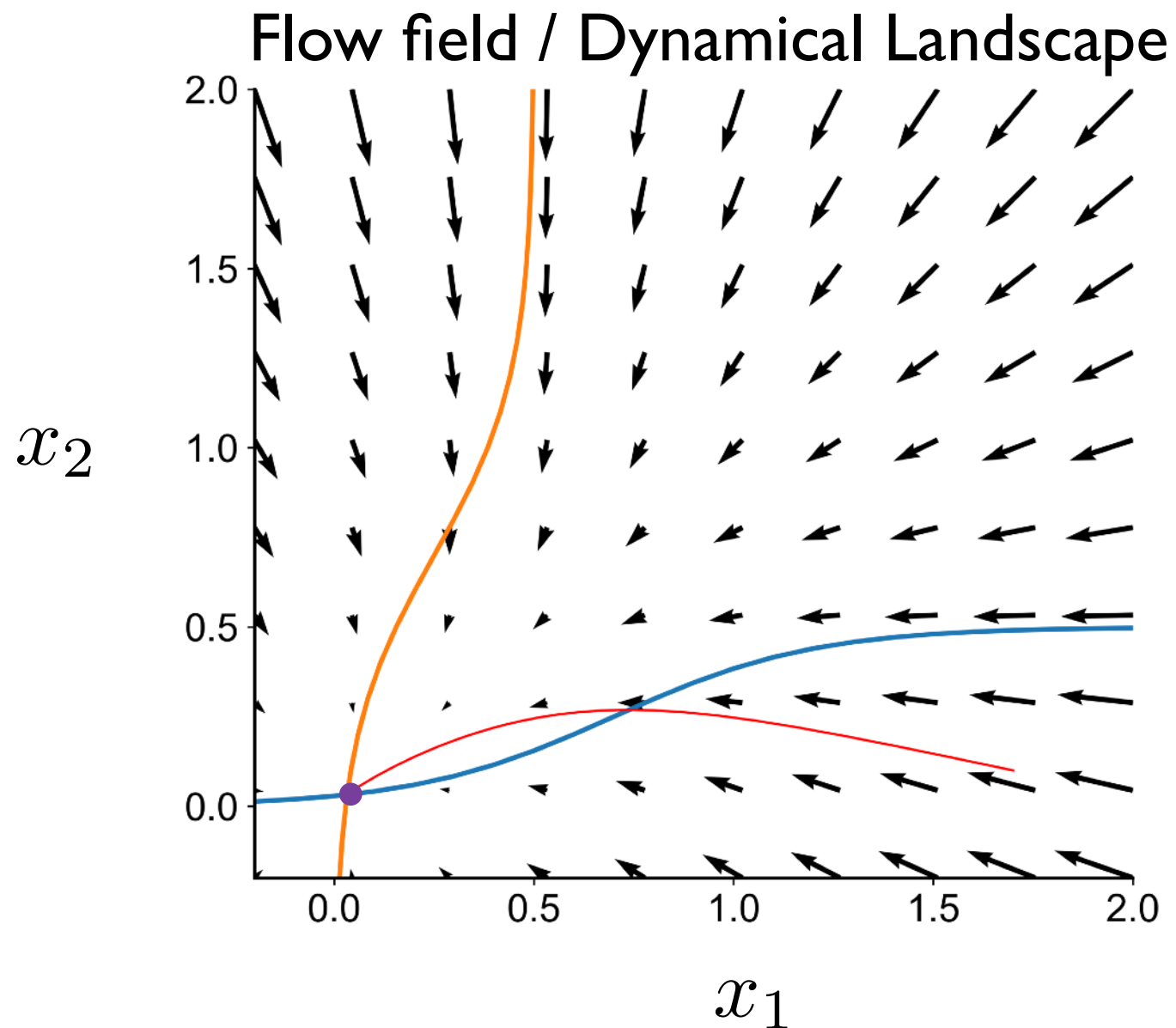
# Two excitatory populations

$$J > 0$$



$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$

$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$

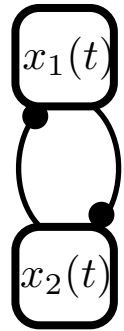


$$\left( \frac{d\mathbf{x}_1}{dt}, \frac{d\mathbf{x}_2}{dt} \right)$$

$$\mathbf{x}(\mathbf{t}) = (x_1(t), x_2(t))$$

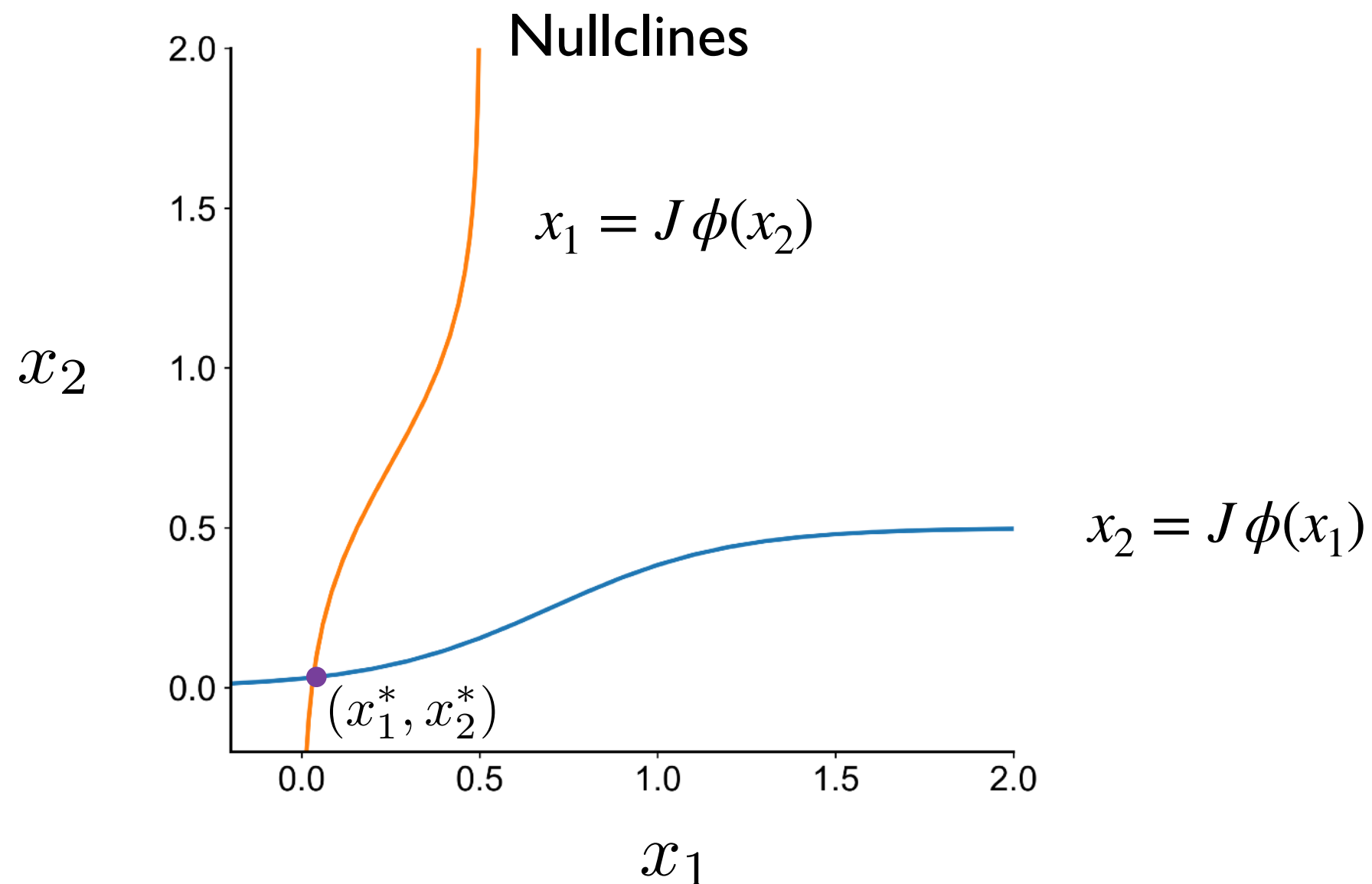
# Two excitatory populations

$$J > 0$$



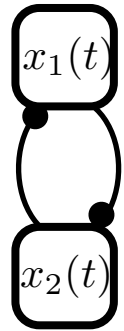
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$

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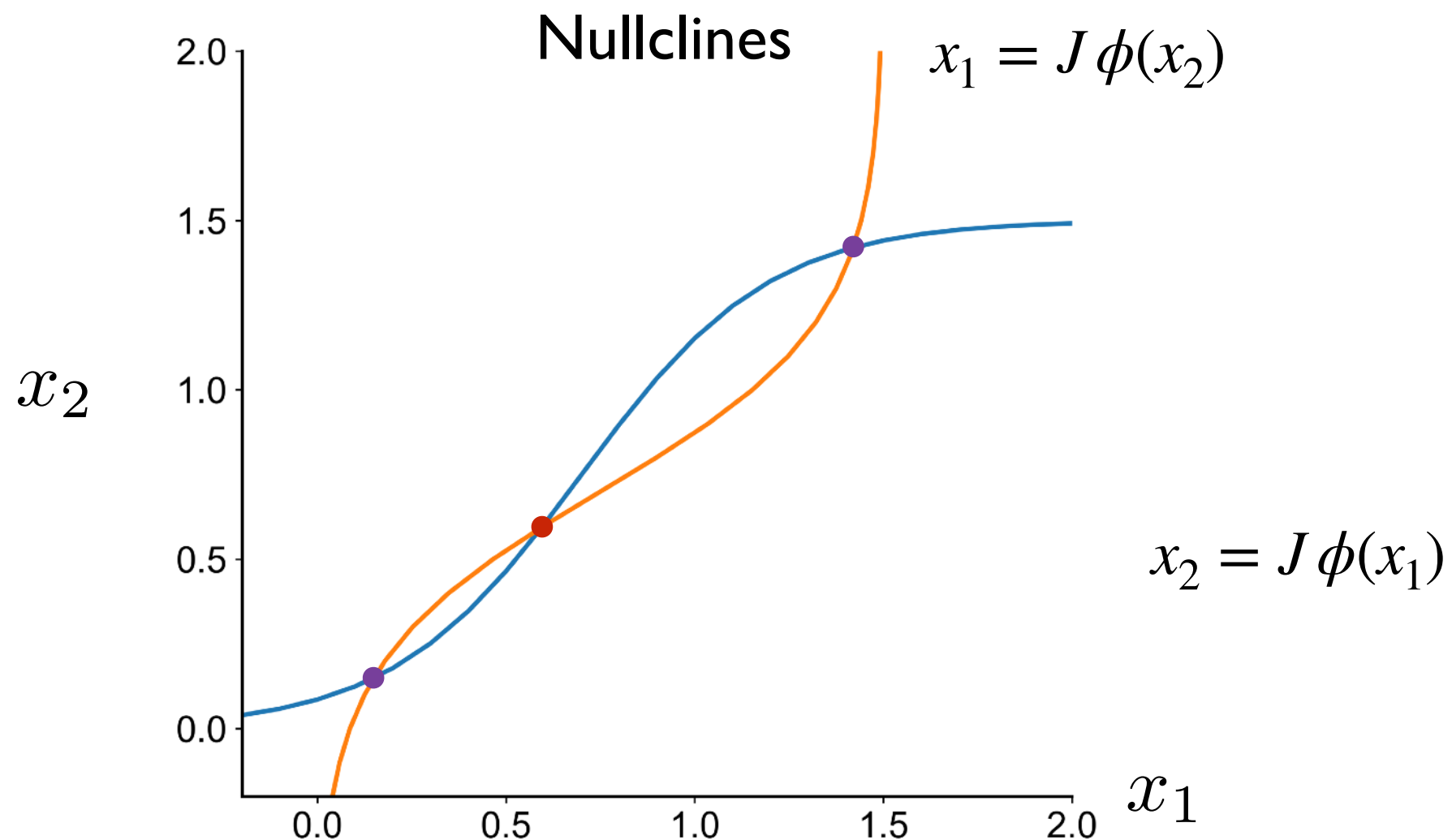
# Two excitatory populations

$$J > 0$$



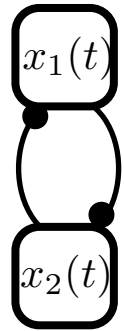
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$

$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$

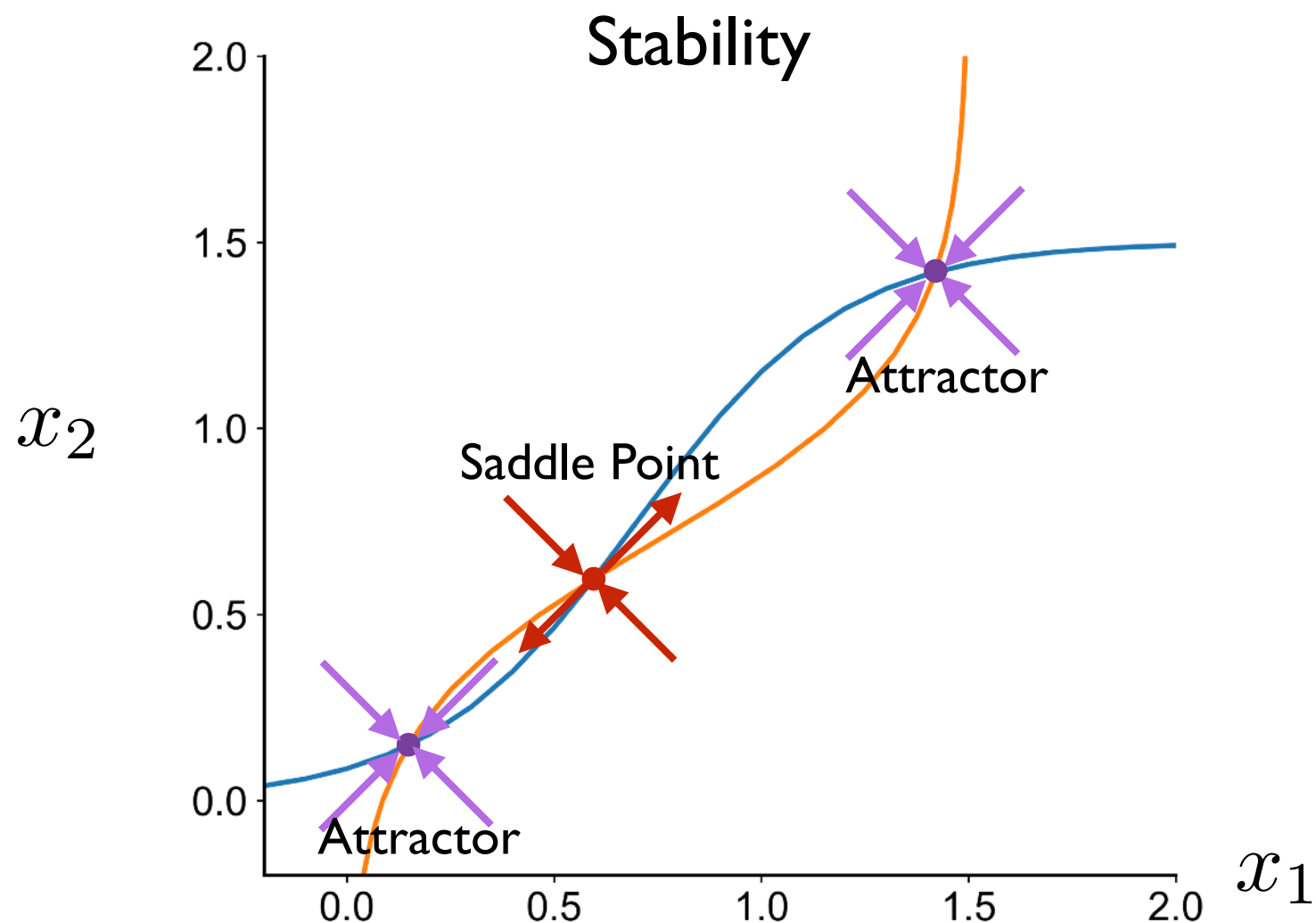


# Two excitatory populations

$$J > 0$$



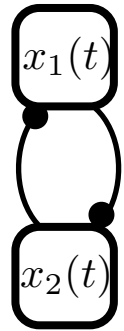
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$



Bi-stability: low vs high activity state

# Two inhibitory populations

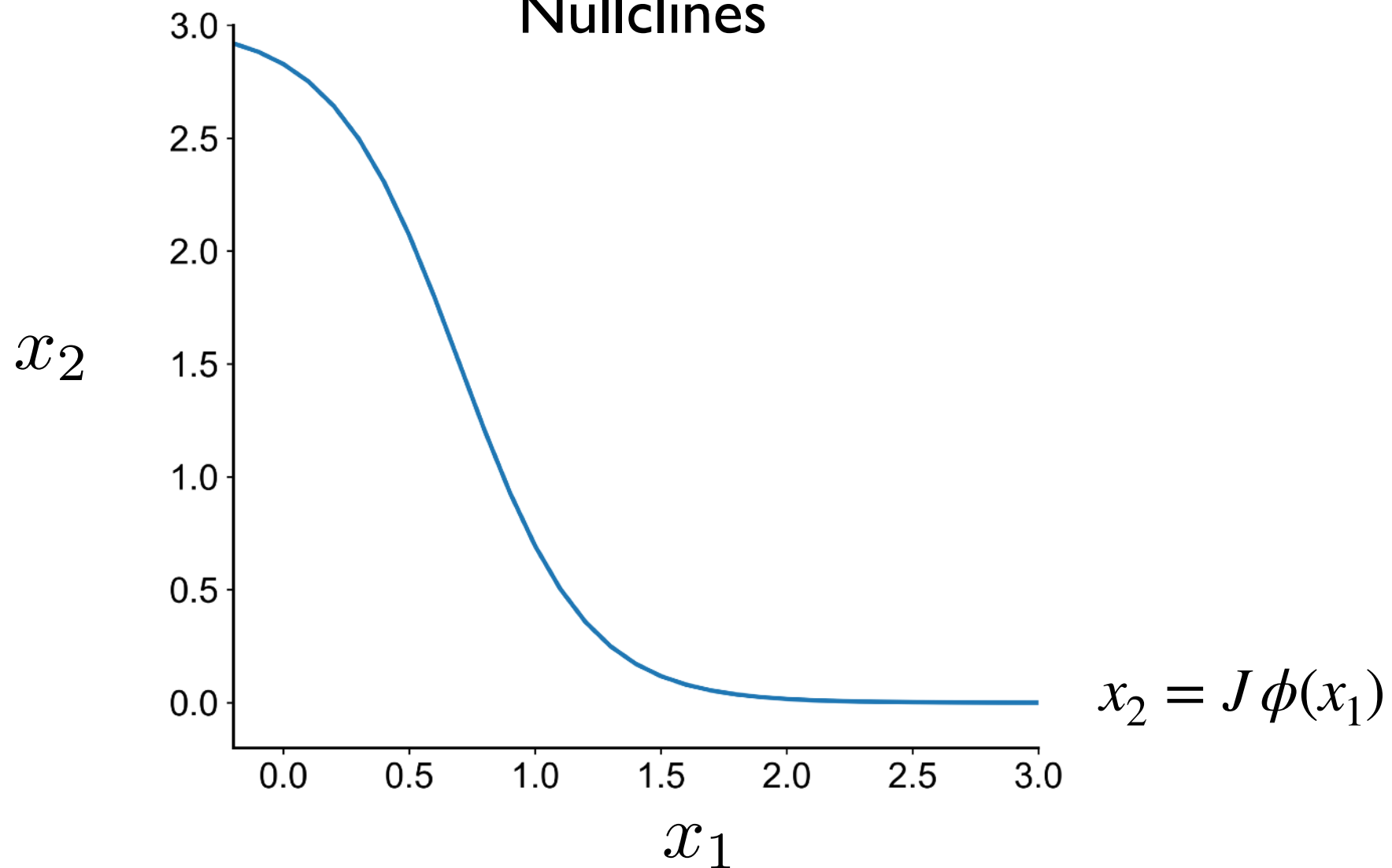
$$J < 0$$



$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$

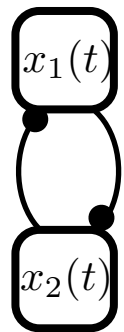
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$

Nullclines

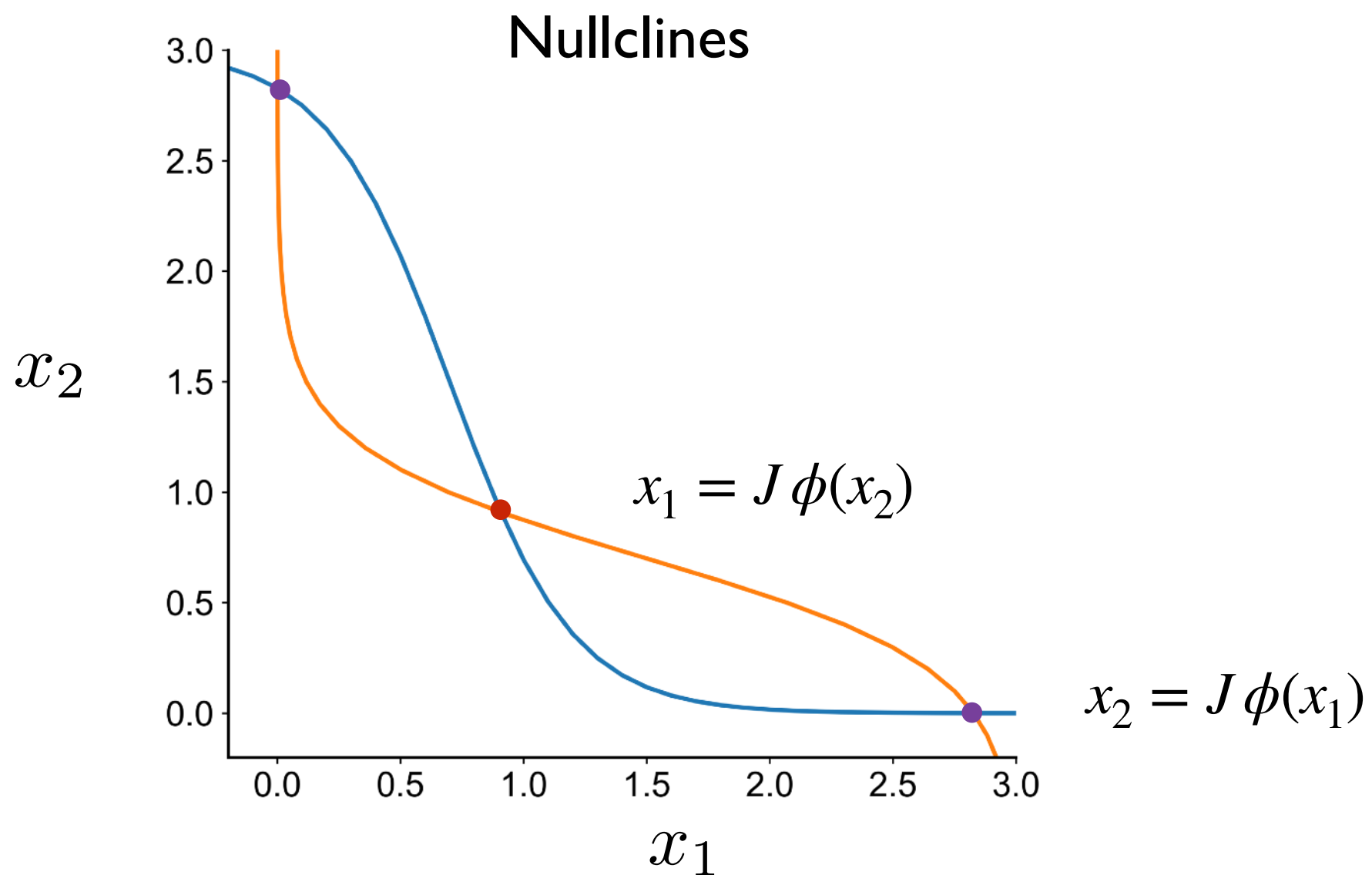


# Two inhibitory populations

$$J < 0$$

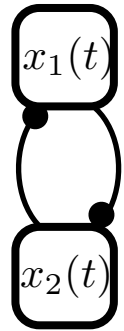


$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$

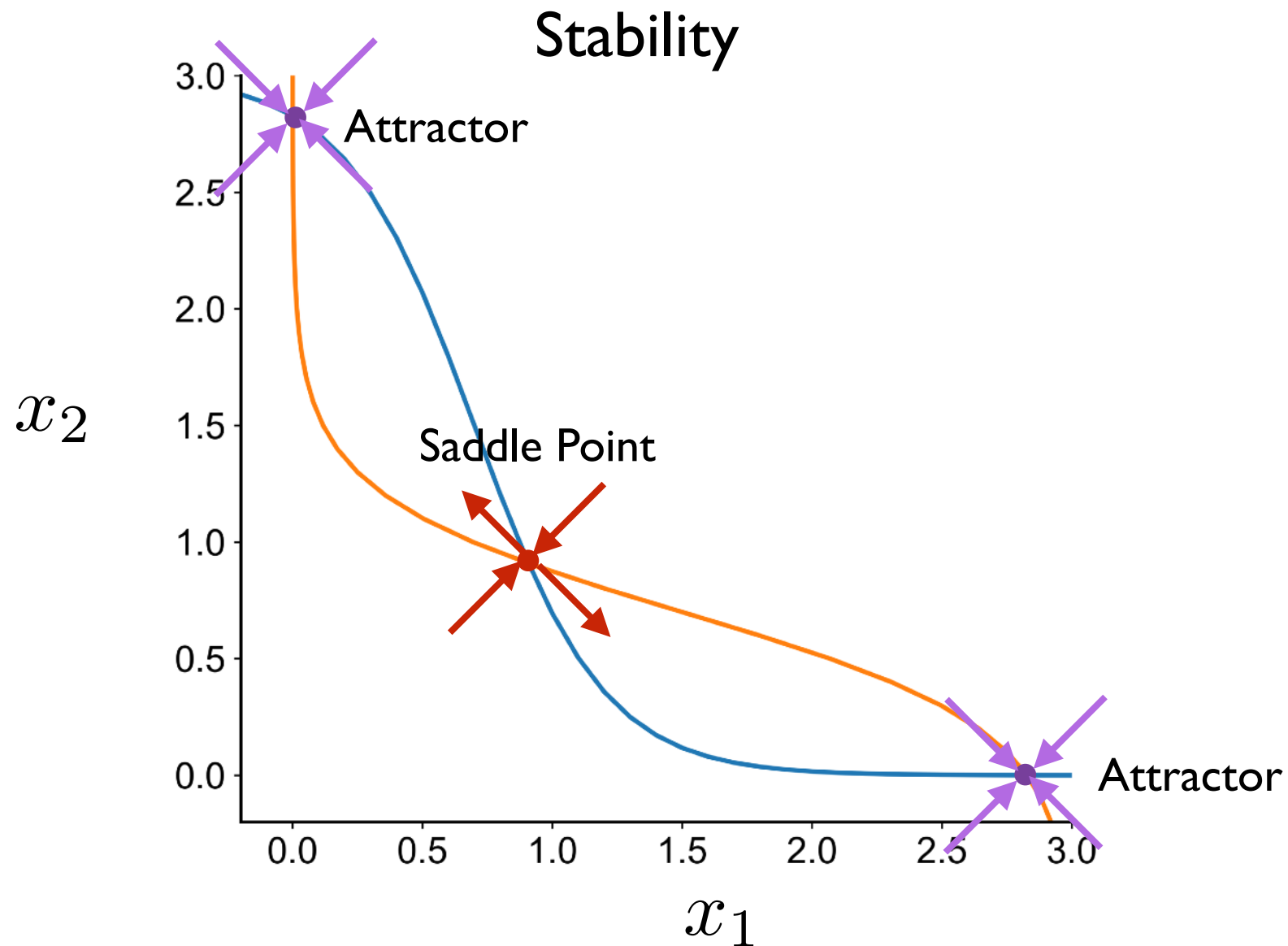


# Two inhibitory populations

$$J < 0$$

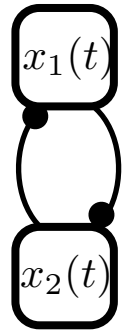


$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$



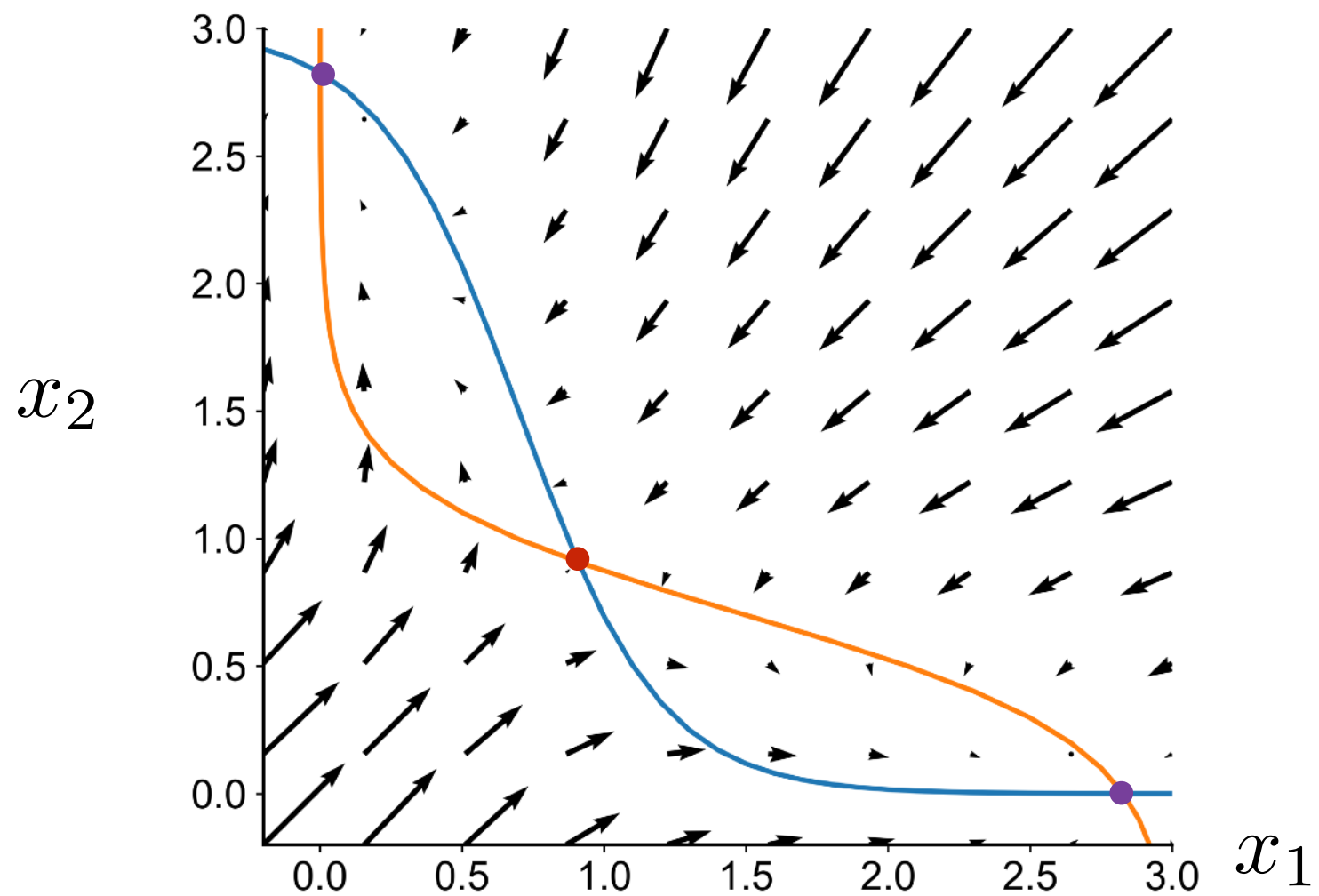
# Two inhibitory populations

$$J < 0$$



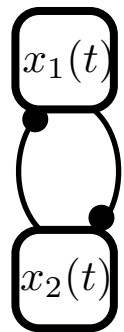
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$

Flow-field



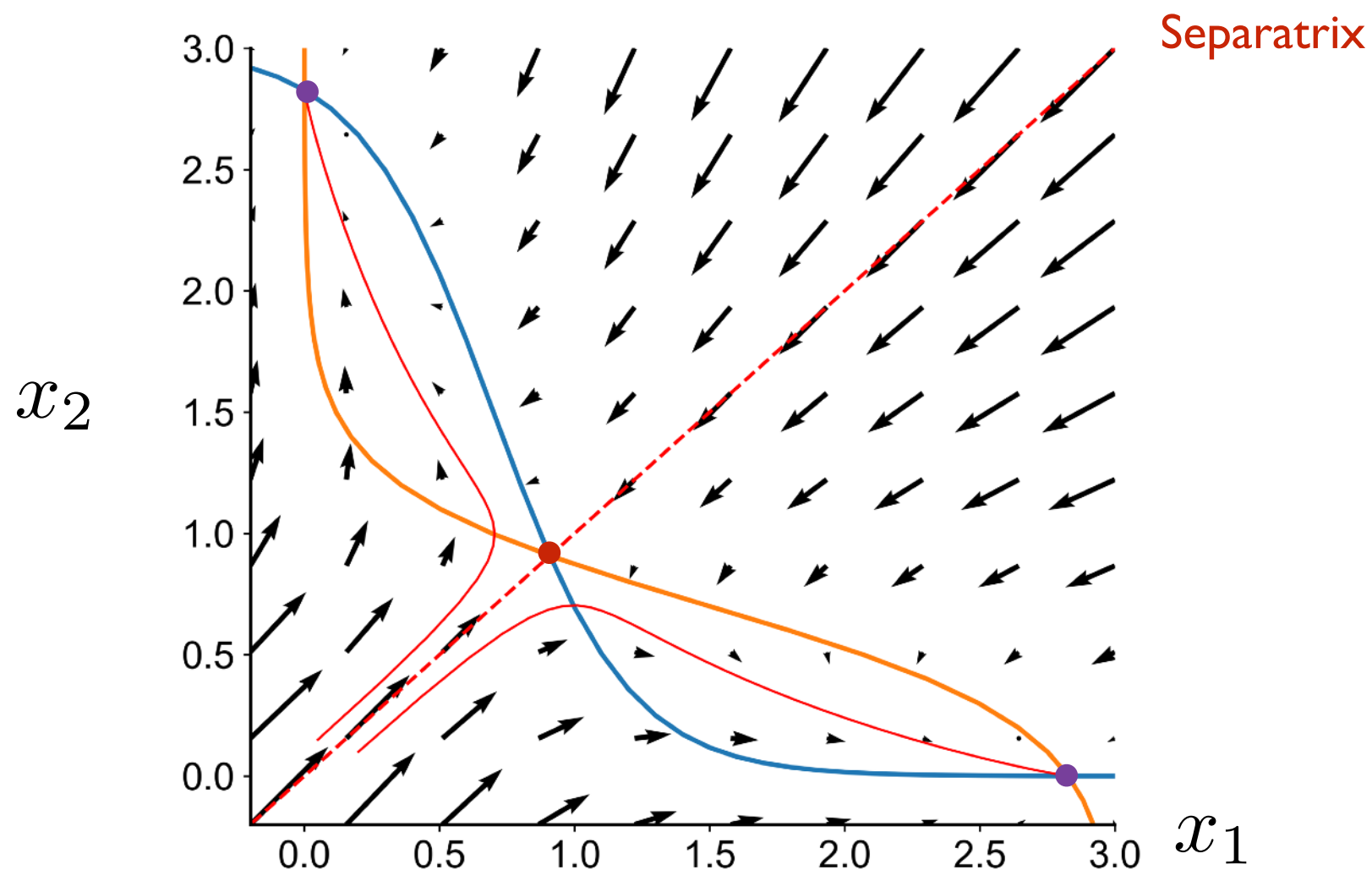
# Two inhibitory populations

$$J < 0$$



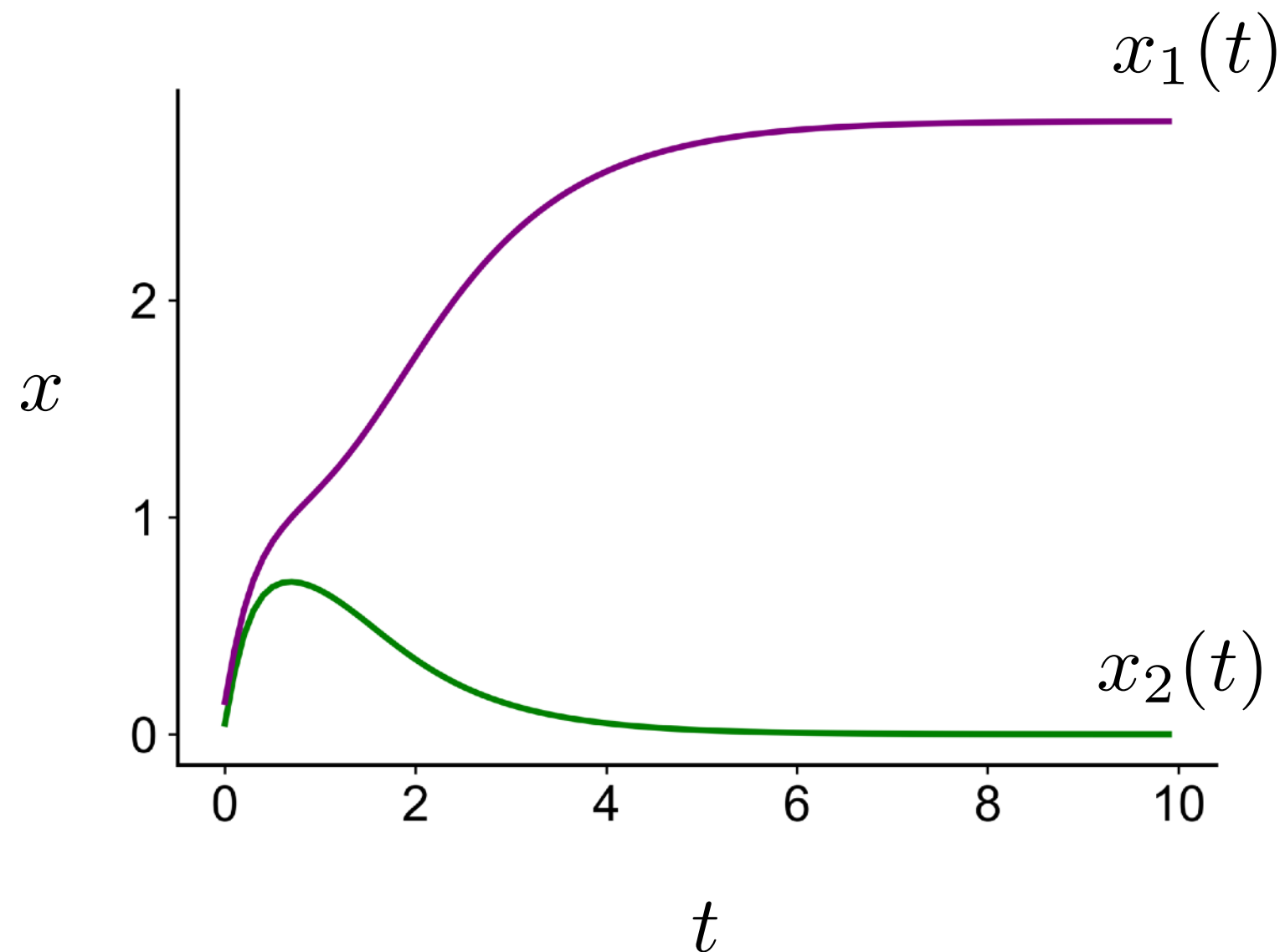
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2)$$
$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1)$$

Flow-field



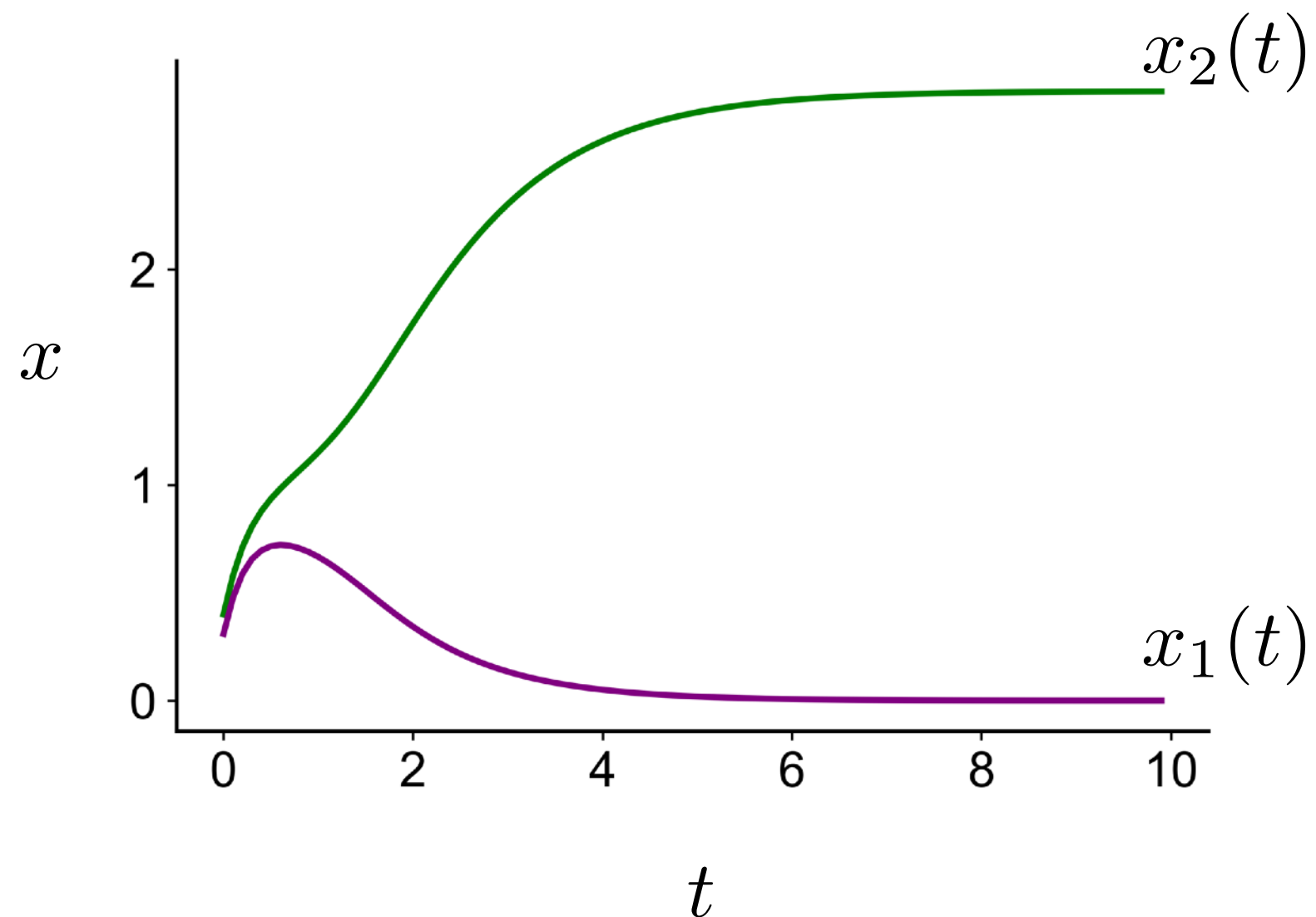
Winner-take-all competition

# Winner - take - all



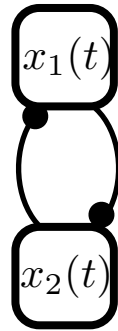
Final state depends on initial condition

# Winner - take - all



Final state depends on initial condition

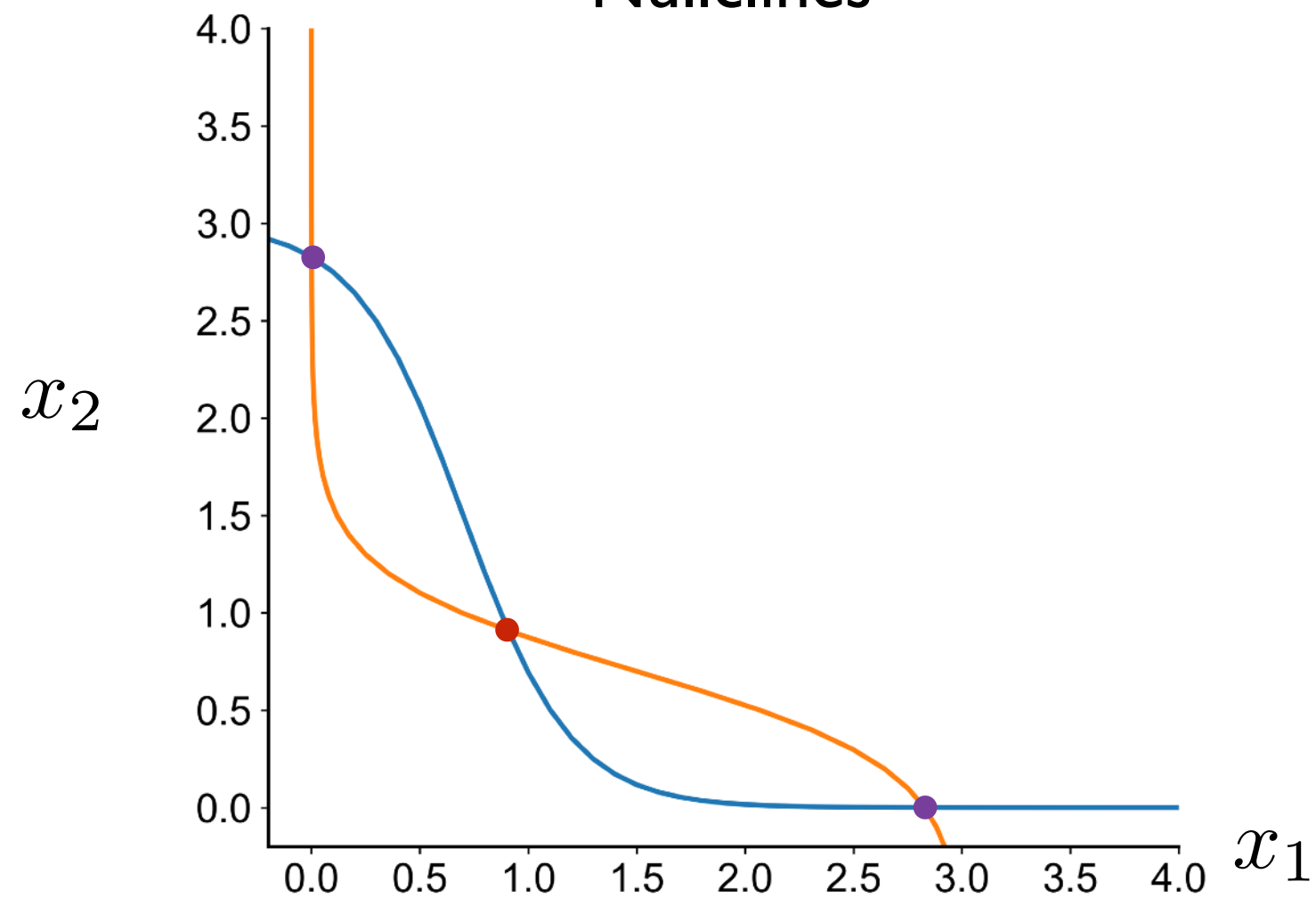
# Effects of inputs



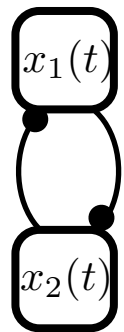
$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2) + I_1^{(ext)}$$

$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1) + I_2^{(ext)}$$

## Nullclines

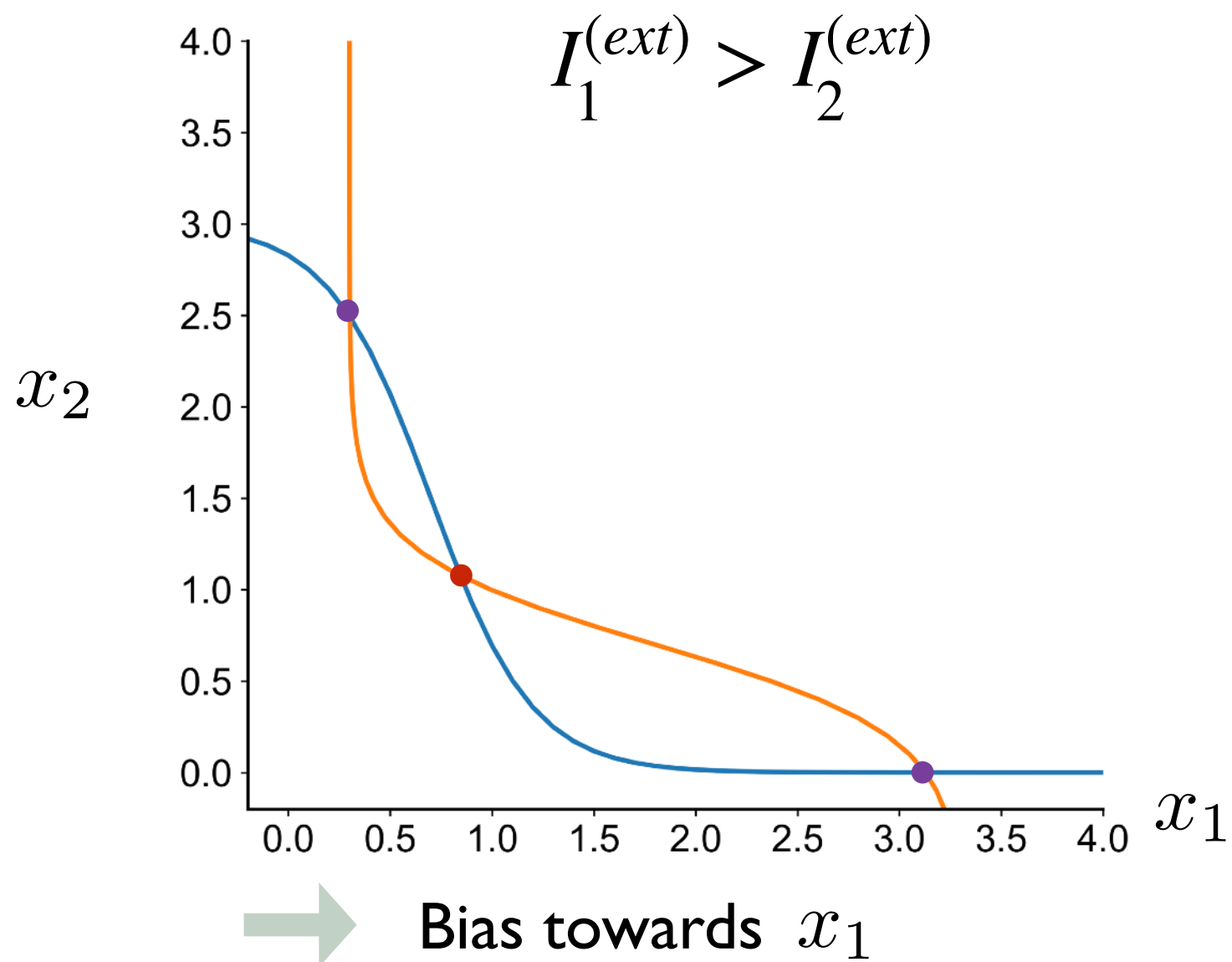


# Effects of inputs

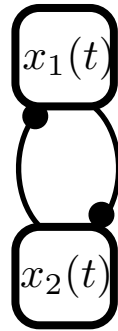


$$\tau \frac{dx_1}{dt} = -x_1 + J \phi(x_2) + I_1^{(ext)}$$

$$\tau \frac{dx_2}{dt} = -x_2 + J \phi(x_1) + I_2^{(ext)}$$

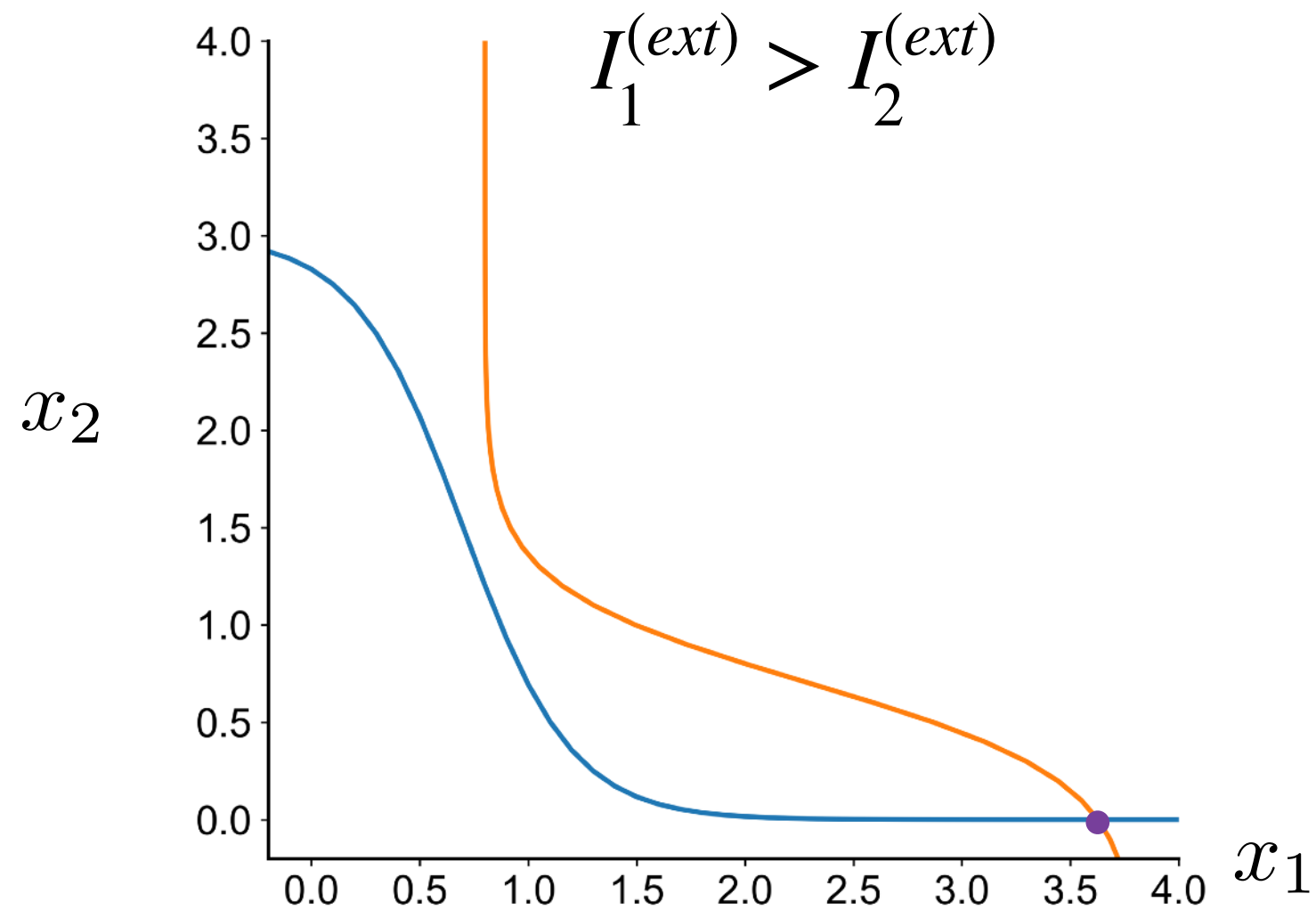


# Effects of inputs



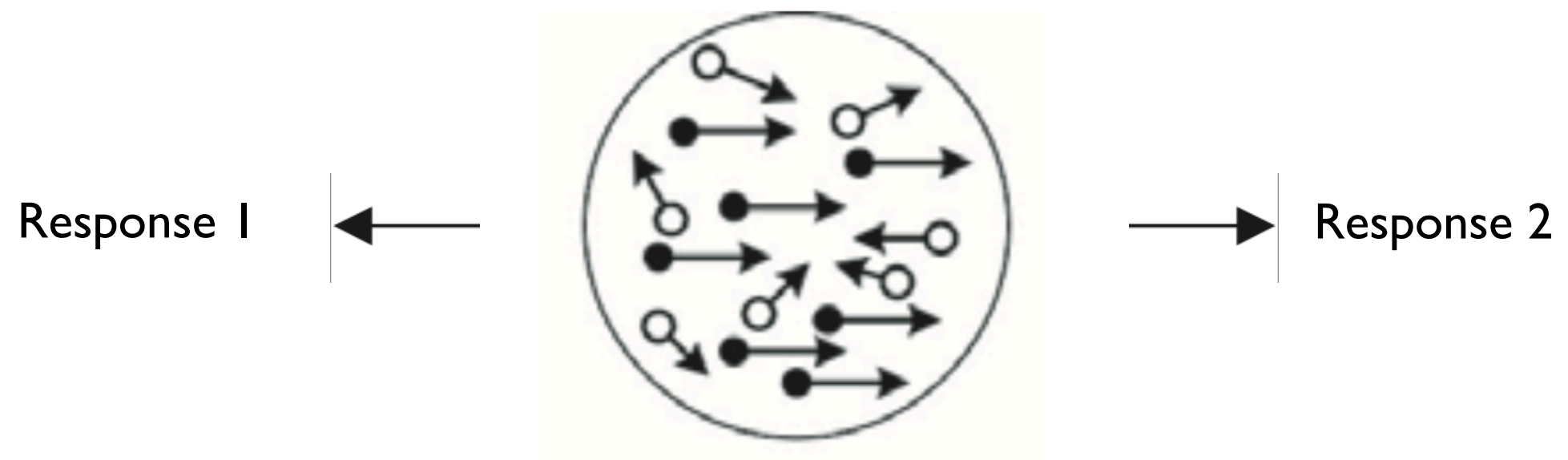
$$\tau \frac{dx_1}{dt} = -x_1 + J\phi(x_2) + I_1^{(ext)}$$

$$\tau \frac{dx_2}{dt} = -x_2 + J\phi(x_1) + I_2^{(ext)}$$



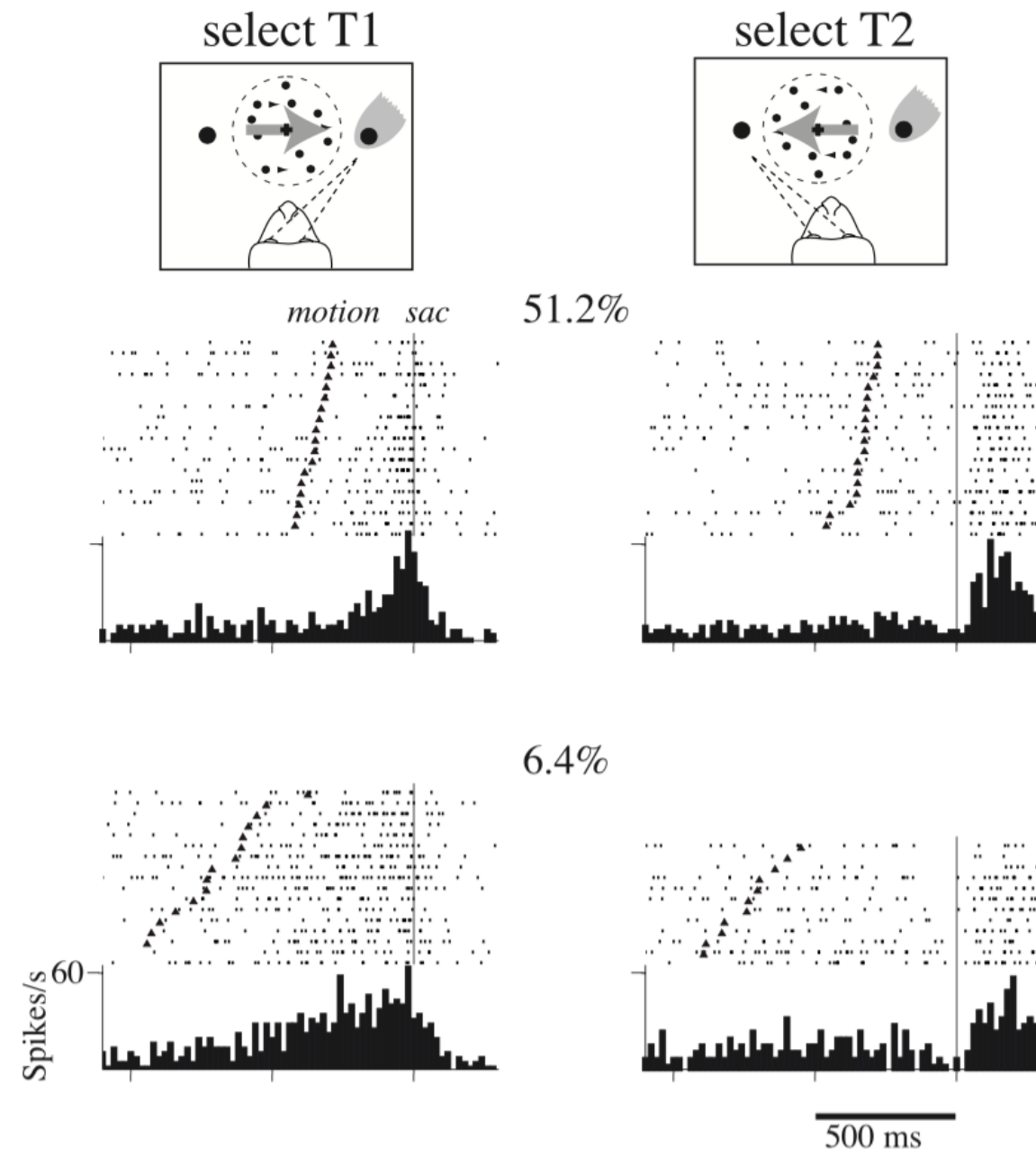
# From neural activity to behaviour

## Perceptual decision-making



# From neural activity to behaviour

## Perceptual decision-making

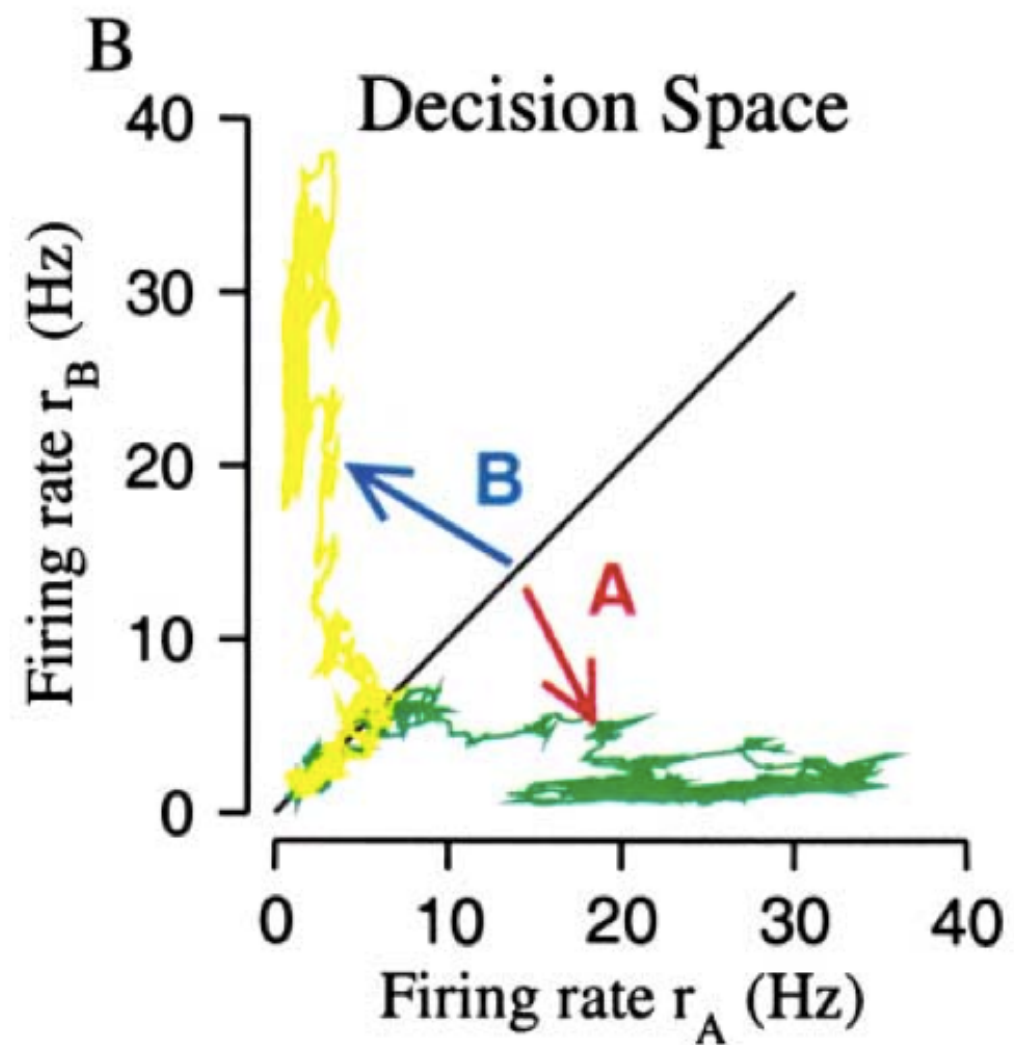


[Gold and Shadlen 2007]



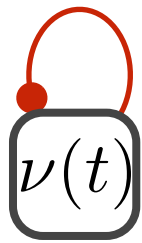
decision= winner-take-all competition between alternatives

# Attractor model of decision making

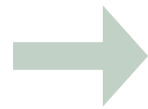


[Wang 2002]

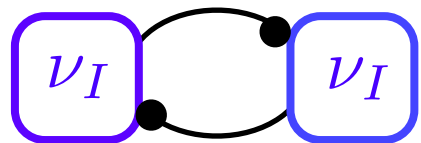
# Summary: basic network mechanisms



Positive feedback



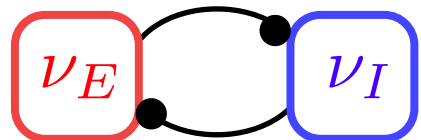
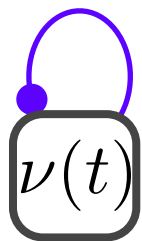
bi-stability, memory, persistent activity



Mutual inhibition



competition, winner-take-all



Delayed inhibition



oscillations

# Recurrent networks

$$\tau \frac{dx_i}{dt} = -x_i + \sum_{j=1}^N J_{ij} \phi(x_j) + I_i^{(ext)}(t)$$

# Outline

1. Single Unit
2. Two Units
3. **Many Units**

# Random recurrent networks

$$\tau \frac{dx_i}{dt} = -x_i + \sum_{j=1}^N J_{ij} \phi(x_j)$$

$J_{ij}$  : Gaussian, mean zero, sd  $\frac{J}{\sqrt{N}}$

$$\phi(x) = \tanh(x)$$

# Random recurrent networks

$$\tau \frac{dx_i}{dt} = -x_i + \sum_{j=1}^N J_{ij} \phi(x_j) \quad \phi(x) = \tanh(x)$$

$J_{ij}$  : Gaussian, mean zero, sd  $\frac{J}{\sqrt{N}}$

1. Fixed point:  $x_i^* = 0$

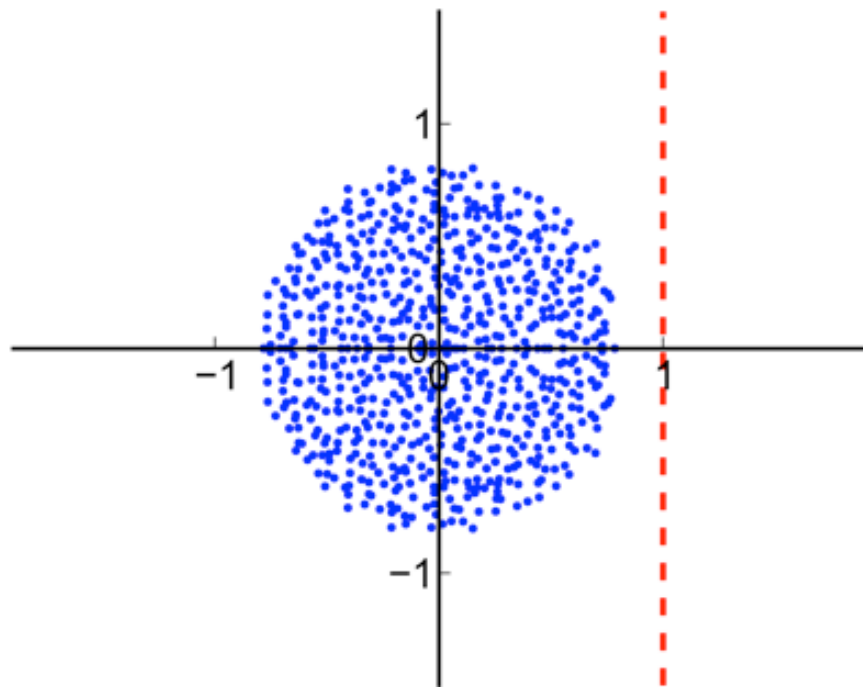
2. Linear stability  $x_i(t) = x_i^* + x_i^{(1)}(t)$


$$\tau \frac{dx_i^{(1)}}{dt} = -x_i^{(1)} + \sum_{j=1}^N J_{ij} x_j^{(1)}$$

# Rate chaos

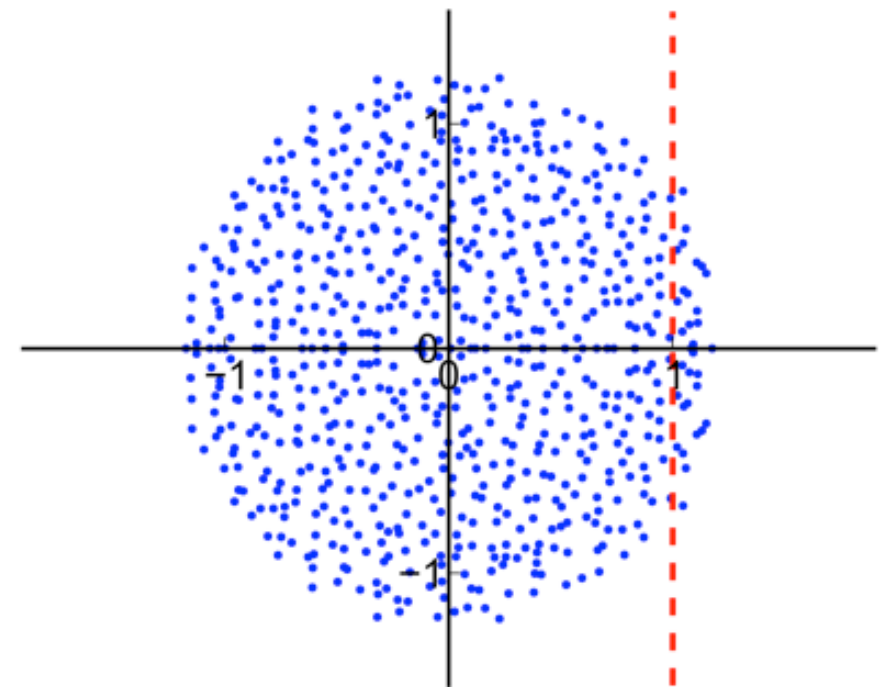
eigenvalues of  $J_{ij}$

$$J < 1$$



fixed point stable

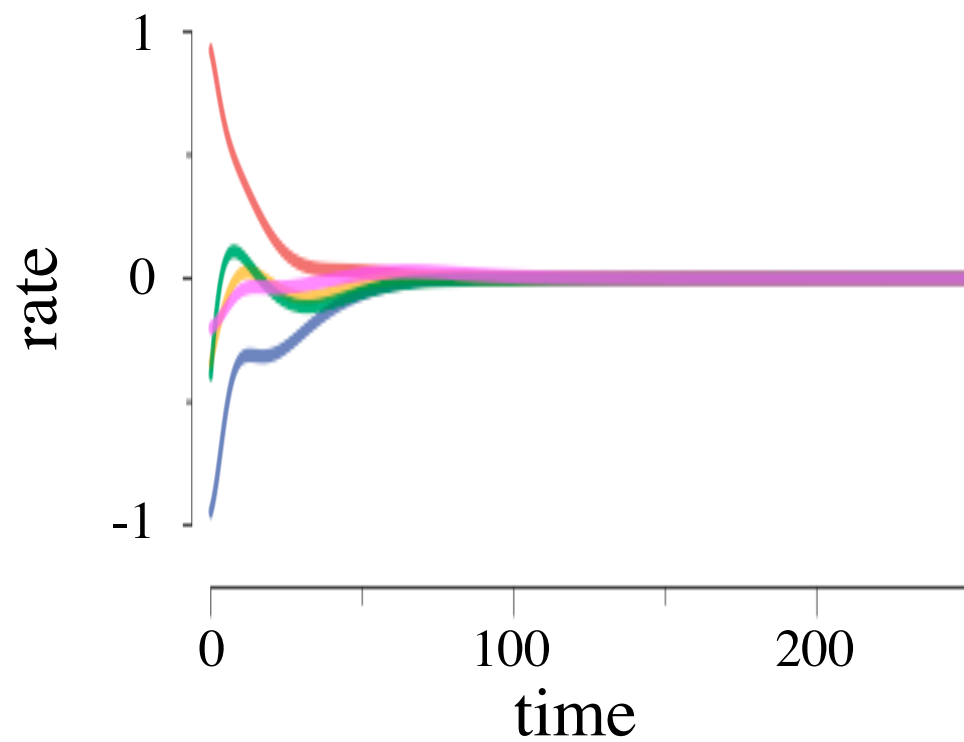
$$J > 1$$



instability to chaos!

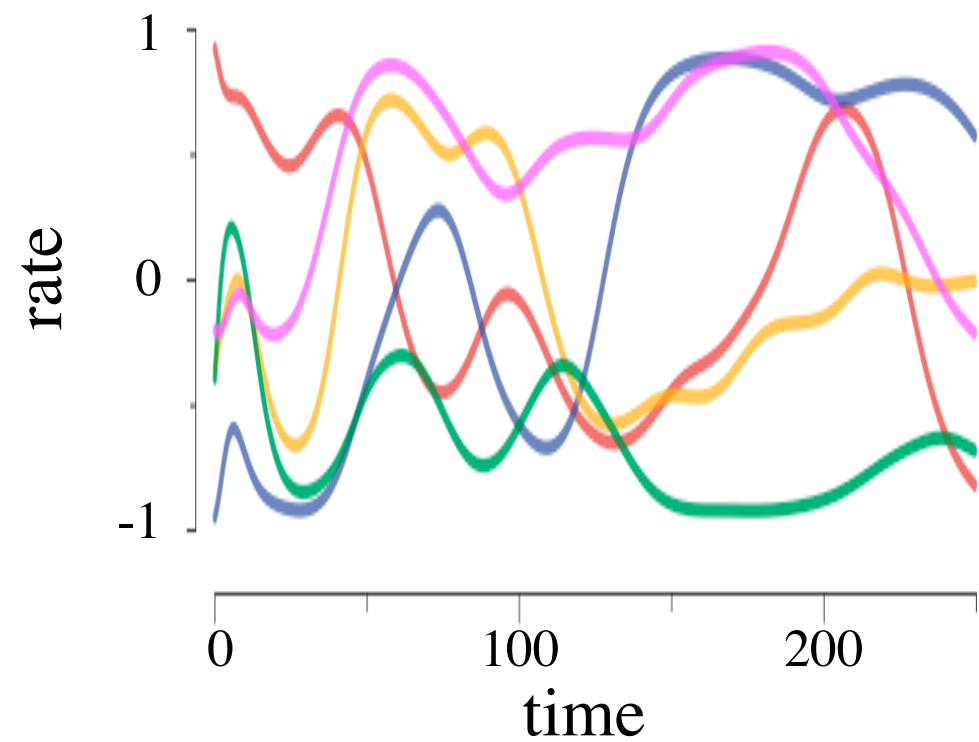
# Rate chaos

$$J < 1$$



fixed point solution

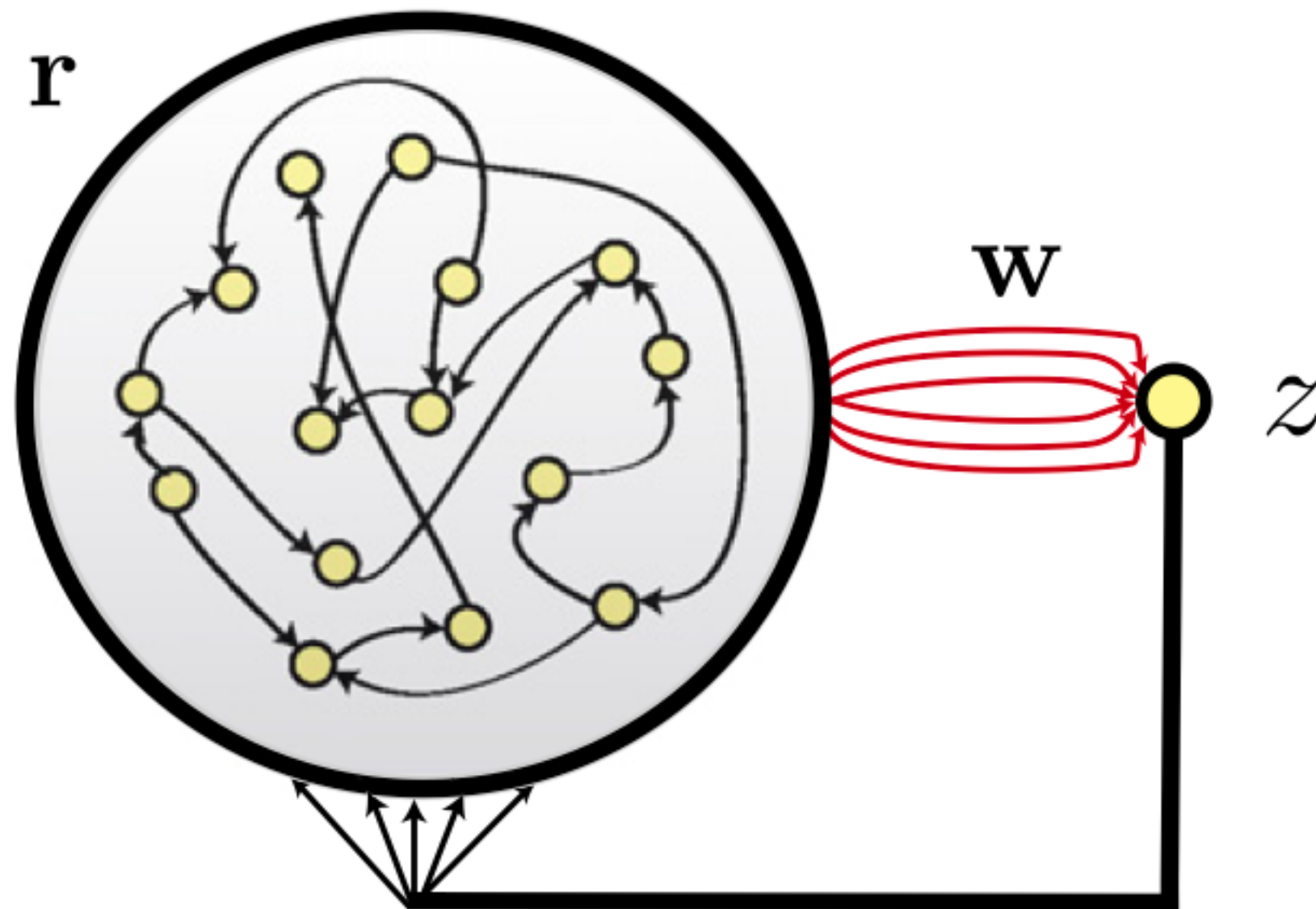
$$J > 1$$



instability to chaos!

# Computational capacity in the chaotic state

Echo-state, Liquid computing, FORCE learning



[Sussillo and Abbott 2009, Toyozumi and Abbott 2011]